The Impact of Parental Resources on Human Capital Investment and Labor Market Outcomes: Evidence from the Great Recession

by

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CES 24-34 **June 2024**

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Abstract

I study the impact of parents' financial resources during adolescence on postsecondary human capital investment and labor market outcomes, using house value changes during the Great Recession of 2007-2009 as a natural experiment. I use several restricted-access datasets from the U.S. Census Bureau to create a novel dataset that includes intergenerational linkages between children and their parents. This data allows me to exploit house value variation within labor markets, addressing the identification concern that local house values are related to local economic conditions. I find that the average decrease to parents' home values lead to persistent decreases in bachelor's degree attainment of 1.26%, earnings of 1.96%, and full-time employment of 1.32%. Children of parents suffering larger house value shocks are more likely to substitute into two-year degree programs, drop out of college, or be enrolled in a college program in their late 20s.

* Email: jakirk2@wisc.edu. I would like to thank Matthew Wiswall, Jeffrey Smith, Jesse Gregory, Christopher Taber, Corina Mommaerts, and Carter Braxton for their invaluable feedback during this project. I am also incredibly grateful to Robert Thomas, for his guidance through the data application process, and for his continued help as I work with the restricted-access U.S. Census Bureau Data. This research was supported by a 2022 Summer Research Fellowship from the University of Wisconsin-Madison. Any views expressed are those of the authors and not those of the U.S. Census Bureau. The Census Bureau has reviewed this data product to ensure appropriate access, use, and disclosure avoidance protection of the confidential source data used to produce this product. This research was performed at a Federal Statistical Research Data Center under FSRDC Project Number 2797. (CBDRB-FY23-P2797-R10886)

1 Introduction

My paper studies the impact of parents' resources on post-secondary human capital investment, earnings, and employment using house value shocks during the Great Recession as a natural experiment. There is a strong, positive relationship between a child's earnings and college education outcomes and their parents' financial resources (e.g. Dynarski and Bailey (2011); Chetty et al. (2014b,c)). Recessions are events that reduce financial resources through income loss and/or reductions in the value of household assets. My paper answers a natural question arising from these facts: what are the intergenerational consequences of recession exposure?

I find that the parents' house values during adolescence have a positive and persistent impact on bachelor's degree attainment, earnings, and full-time employment. The average house value change during the Great Recession in my sample is -15.8%. Exposure to the average house value shock decreases bachelor's degree attainment by 1.26%, annual earnings by 1.95%, and full-time employment of 1.32%. Meanwhile, declines to parents' house values during adolescence increases associate's degree attainment, increases the likelihood of dropping out of college, and increases college enrollment in a child's late 20s. Declines to parents' wealth also increases parental co-residence during adulthood and decreases migration to different labor markets. These results imply that recessions have a wide range of negative long-run impacts due to the intergenerational transmission of recession shocks from parents to their children.

A significant empirical challenge of using the Great Recession as a natural experiment to

study the influence of parents' housing values on college and labor market outcomes is that local house values were strongly related to local economic conditions. In particular, there was a negative relationship between changes to the local unemployment rate and changes to home prices, which means that stronger labor markets were associated with higher housing values. Failing to control for the confounding influence of local labor markets will result in biased estimates of the impact of parents' house values. To highlight this, consider the following stylized regression of bachelor's degree attainment on the change in parents' house values:

$$ba_attainment = \beta \Delta HV + \epsilon \tag{1}$$

Ceteris paribus, the expectation is that an estimate of the impact of parents' house values on bachelor's degree attainment is positive (i.e. $\beta > 0$). However, the estimate will be biased since house value changes, ΔHV, are correlated with local labor market conditions. The influence of local labor markets is part of the error term ϵ and influences *ba_attainment*. It is not immediately clear if the direction of the bias will be positive or negative due to the fact that there are several channels through which local labor markets influence college decisions. Larger (positive) changes to house values are associated with improved labor market prospects for parents, which increase their (non-housing) resources and thus increase bachelor's degree attainment. This will result in a positive bias for the estimate of β. Stronger local labor markets may increase the child's returns to education, which also results in a positive bias of the estimate of β. On the other hand, stronger local labor markets increase the child's opportunity cost of higher education, which has a

downward influence on college investment. This results in a negative bias for the estimate of β.

The relationship between parents' house values and the opportunity cost of education is particularly concerning in the context of the Great Recession. Rinz (2022) finds that the earnings of millennial cohorts were the most impacted by the recession, and that earnings losses were larger for those who lived in labor markets that suffered larger unemployment shocks (and therefore, larger house value shocks) during the recession. The cohorts that suffer the largest earnings impacts overlap with the cohorts that I study in this paper. Furthermore, Barr and Turner (2013) report that aggregate college enrollment increased by 2.8 million during the Great Recession. The large increase in college enrollment is an indication of a strong response to the decrease in the opportunity cost of education.

I overcome the empirical challenge posed by the relationship between house values and local labor markets by estimating a generalized difference-in-differences specification. A key feature of the model is that it includes metropolitan area-by-cohort fixed effects to non-parametrically control for local labor market conditions. My specification exploits *within* labor market variation in housing values, so that the variation in parents' house values is arguably like-random. It is not possible to implement this research design with commonly known panel survey datasets such as the Panel Survey of Income Dynamics (PSID) or the National Longitudinal Survey of Youth (NLSY) because these data have insufficient sample sizes. Therefore, I create a novel dataset that links restricted-access microdata from the U.S. Census Bureau. I create this dataset by linking the American Community Survey (ACS), the 2000 Decennial Census, and the Master Address File Auxiliary Reference File (MAF-ARF).

My data contains four key elements that I need to implement my research design. It has intergenerational linkages between parents and children, and has information on children's outcomes observed during adulthood. It also includes precise geographic information during the Great Recession, which allows me to determine the magnitude of house value shocks experienced by households. Finally, it yields an analysis sample that is large enough to control for local economic conditions using metropolitan area-by-cohort fixed effects.

The prior literature that uses changes to housing wealth as a natural experiment to study the impact of parents' resources on children's education and labor market outcomes does not control for local labor market conditions with metropolitan area-by-cohort fixed effects (Lovenheim, 2011; Lovenheim and Reynolds, 2013; Cooper and Luengo-Prado, 2015). In this paper, I show that estimates from an approach that does not exploit within labor market variation in house values are smaller than estimates from my preferred specification. This suggests the association between local house values and labor markets creates negative bias when studying the influence of parents' house values on college attainment.

1.1 Related Literature

My paper contributes to three strands of the economics literature; the literature studying the impact of parents' financial resources on post-secondary human capital investment, the literature studying the labor market scarring effects of exposure to recessions, and the literature studying the influence of recessions on post-secondary human capital investment.

Several papers use "natural experiments" that exploit plausibly exogenous shocks to parents' resources to study the impact of parents' financial resources on their children's college

investment. Hilger (2016) studies the impact of parents' job loss on college enrollment and earnings and finds a small impact of job loss on enrollment of less than 0.5 percentage points (p.p.). There are several papers that exploit policy variation in the generosity of the Earned Income Tax Credit (EITC) to study the impact of parents' income on college attendance, attainment, and earnings (Chetty et al., 2011; Bastian and Michelmore, 2018; Manoli and Turner, 2018). These papers find that increases in income from the EITC lead to substantial increases in college attendance, degree attainment, and earnings during adulthood. Bulman et al. (2021) studies the impact of household lottery wins on college attendance and finds small enrollment effects. The effects uncovered in these papers are localized to a specific portion of the income distribution (EITC recipients) or specific populations (children of parents suffering job loss or lottery winners). My paper contributes to this literature by studying a resource shock that is salient to a broader population, namely the children of homeowners.

There are also several papers that use home value variation as a natural experiment to study the relationship between parents' housing wealth and children's college education and labor market outcomes. These papers find that housing wealth increases college enrollment (Lovenheim, 2011), college degree completion among students who attend college (Lovenheim and Reynolds, 2013; Hotz et al., 2023) and future earnings (Cooper and Luengo-Prado, 2015). Hotz et al. (2023) show that parents' use their housing wealth to help their children finance investments in college education. These papers use data from the Panel Study of Income Dynamics (PSID) (Lovenheim, 2011; Cooper and Luengo-Prado, 2015; Hotz et al., 2023) or the National Longitudinal Survey of Youth (NLSY) (Lovenheim and Reynolds, 2013). An implication of using these data is analysis samples that are too small to facilitate the

inclusion of fixed effects needed to exploit within labor market variation in house values. Instead, these papers use parametric controls to account for the relationship between local house price changes and local labor markets. I make a pair of contributions to the existing literature. My paper is the first to study the impact of house value changes during the Great Recession, a context that is substantially different than previous cohorts studied[1]. Additionally, I provide suggestive evidence that using empirical methods that do not use metropolitan area-by-cohort fixed effects to control for local labor markets yields biased estimates in the context of the Great Recession.

This paper is also related to the literature studying the labor market scarring effects of recessions. This literature focuses on recessions' impact on those who are already in the labor market (Yagan, 2019; Rinz, 2022) or on new labor market entrants (Kahn, 2010; Oreopoulos et al., 2012; Altonji et al., 2016; Schwandt and von Wachter, 2019; Rothstein, 2020; Anstreicher and Miller, 2023). The main finding from this literature is that exposure to recessions leads to persistently lower earnings. My results imply that the scarring effect of recessions is not limited to those already attached to or entering the labor market. I show that the intergenerational transmission of recession shocks from parents to their children persistently reduces the future education attainment, earnings, and employment of children.

My paper is closely related to Stuart (2022), who studies the impact of the double-dip recession of the early 1980s on individuals who were children at the time of the recession. This paper finds that recession exposure during early childhood significantly decreased long-run education attainment and earnings, but that exposure during adolescence did not have

[1]Hotz et al. (2023) includes some individuals who turned 18 during the beginning of the Recession, but the majority of the sample turns 18 during or prior to the 2000s housing boom.

long-run impacts. In contrast, I do find that shocks to parents' resources during adolescence reduced future bachelor's degree attainment and earnings in the context of the Great Recession. It is not surprising that the results of my paper and Stuart (2022) differ. The inflation adjusted price of college attendance more than doubled between the start of the 1980s recession and the Great Recession[2], which means that parents' resources likely became a more important factor in the financing of college. This logic is consistent with the existing literature showing that the relationship between parents' financial resources and the college education investments of their children has grown stronger since the 1980s (Belley and Lochner, 2007; Dynarski and Bailey, 2011).

My paper is also related to the literature that studies the relationship between the business cycle and post-secondary human capital investment. This literature finds evidence of a countercyclical relationship between investment in higher education and the business cycle. (Betts and McFarland, 1995; Sakellaris and Spilimbergo, 2000; Dellas and Sakellaris, 2003; Kahn, 2010; Blom et al., 2021). It also finds evidence that the Great Recession had a particularly strong impact on increasing college enrollment (Barr and Turner, 2013; Long, 2015). I contribute to this literature by providing evidence that the countercyclical impact of recessions on higher education investments is understated because shocks to parents' resources depress bachelor's degree attainment.[3]

[2]The average annual cost of tuition, fees, room, and board at a four-year college was $11,393 in 1980-1981 at the beginning of the 1980-1982 recession. That same figure increased to $25,797 in 2007-2008 at the beginning of the Great Recession. Dollar amounts are adjusted for inflation and reported in constant 2021-2022 dollars. Data from the National Center of Education Statistics Digest of Educational Statistics, Table 330.10. `https://nces.ed.gov/programs/digest/d22/tables/dt22_330.10.asp`.

[3]There are a couple of papers that consider the role that credit constraints play in determining enrollment during business cycle fluctuations, but not attainment. Sakellaris and Spilimbergo (2000) find that enrollment of foreign students from OECD countries in U.S. universities is counter-cyclical, but that it is procylical for students from non-OECD countries. Christian (2007) does not find evidence that the cyclicality of

2 Background

2.1 Theoretical Issues Concerning House Value Changes and Investment in Higher Education

Credit Constraints

There are several possibilities that explain why a decrease to parents' ability to finance their child's investment in college education could decrease bachelor's degree attainment. A decrease to parents' resources could reduce education attainment by imposing credit constraints that prevent optimal investment[4] in higher education (e.g. Lochner and Monge-Naranjo (2011, 2012)). Black et al. (2020) provide evidence that reducing credit constraints is a plausible channel through which parents' resources increased education attainment during and in the aftermath of the Great Recession. The authors find that credit constraints reduced bachelor's degree attainment prior to the expansion of federal student loan borrowing limits in 2007 and 2008. They also show that there is still a large amount of bunching that occurs at new federal loan limits, which means that it is possible that some prospective students still faced borrowing constraints that reduce college graduation in the aftermath of the recession.

Income Risk and Education as a Normal Good

enrollment differs by students from home-owning vs. non-home-owning households. Christian (2007) does find that enrollment from low income households is more *pro-cyclical*, indicating that credit constraints could be relevant for lower income households.

[4]Optimal investment means that child invests in higher education as long as the returns to human capital investment are greater than or equal to the economic cost of the investment (e.g. Becker (1962)).

Additionally (or alternatively), decreases to parents' resources could decrease degree attainment if their children are risk averse and parents' resources reduce the income risk associated with investing in higher education. Students' future income is uncertain when they attend college, and many finance their higher education investments using student loans that must be repaid regardless of future income. Risk averse (potential) students might not invest in a bachelor's degree, even if the expected monetary returns are greater than the cost of education. Mumford (2022) shows that enrolling in an income share agreement that reduces the income risk of investing in higher education increases degree completion. Parents' financial resources could increase education investment by helping their children pay for college, and hence reducing the amount of income risk their children are exposed to by borrowing. Finally, decreases to parents' resources could decrease degree attainment if households treat college as a normal consumption good (Bulman et al., 2021). Determining the exact channel through which changes to parents' resources during adolescence impact college attainment is beyond the scope of this paper.

Parents' House Values and College Financing

Changes to the value of a house are relevant resource shocks that could influence the ability of household to finance a child's investment in higher education. Households internalize local house value changes as real changes to the value of their housing asset (see Appendix Section A.3). Household borrowing and consumption responds to changes to a house's value (e.g. Mian and Sufi (2011); Aladangady (2017)). Mian et al. (2013) estimate that the marginal propensity to consume out of housing wealth was between \$0.05-\$0.07 per \$1 of

housing wealth during the Great Recession. During the recession, negative shocks to house values decreased home equity, which in turn could have prevented households from borrowing against the value of their home to finance their child's investments in higher education. In addition to impacting a household's ability to borrow against their home equity, decreases in the value of a house reduces a household's permanent income if they were planning on selling their home in the near future. In this case, the reduction in permanent income could reduce the amount that households spent on their child's college education.

Parents' housing wealth is not taken into account when determining the amount of federal financial aid that a child is eligible to receive. This is despite the fact that a change to the value of household's housing asset can alter the amount of financial support that parents give to their children to finance college. Therefore, decreases to home equity during the Great Recession were not offset with increases in financial aid. In contrast, other shocks to resources such as income reductions resulting from job loss does impact a family's expected contribution to their child's education, and hence the amount of federal financial aid that they are eligible for.

2.2 Housing Value Changes During The Great Recession

The Great Recession of 2007-2009 was the largest recession in the United States since World War II. The recession began with a slowdown in consumer spending and residential investment that followed a period characterized by large increases in household debt. The recession was amplified by a financial crisis that began with the collapse of Lehman Brothers in September 2008. One of the prominent characteristics of the Great Recession is that it

was preceded by the 2000s housing bubble in which there was a sharp increase in housing values and homeownership beginning in the early 2000s. This was followed by the collapse of the U.S. housing market, in which house values declined by $5.5 trillion (Mian and Sufi, 2014). The collapse of the housing market during the recession motivates using local house values as a proxy to study the impact of parents' resources on the future education and labor market outcomes of their children.

Mian and Sufi (2009) highlight several unique facts about the 2000s housing boom and subsequent bust during the Great Recession. The acceleration in local housing prices during the 2000s was not driven by an underlying improvement in local economic conditions. In fact, the correlation at the ZIP code level between income growth and both house price growth and mortgage credit origination was slightly *negative* between 2002 and 2005. Instead of being driven by economic fundamentals, the increase in mortgage credit was concentrated in ZIP codes that had ex ante high rates of borrowers with poor credit ratings. The increase in the supply of credit to subprime borrowers was responsible for the rapid increase in new debt, and perhaps the increase in housing prices itself. These "subprime" ZIP codes that experienced large increases in credit before the recession were the same ZIP codes that had high rates of mortgage defaults once the housing bubble popped. An important takeaway from these facts is that changes to average ZIP code house values within a metropolitan area are not primarily driven by differential recession shocks across the labor market. Instead, they were driven by the ex ante creditworthiness of residents.

Unsurprisingly, the fact that large house value shocks within labor markets occurred in "subprime" ZIP codes means that there are baseline differences in outcomes between ZIP

codes experiencing different house value changes during the recession (see Appendix Figure C.3). The fact that house value changes within labor markets were driven by the ex ante characteristics of ZIP codes informs the research design that I implement. I use ZIP code house value changes to proxy for changes to parents' house values in a generalized difference-in-differences framework. My specification includes ZIP code fixed effects that control for baseline initial differences in outcomes between ZIP codes.

ZIP code level housing values decreased by 17.5% on average between January 2007 and December 2009, with homes losing an average nominal value of $57,300[5]. My research design leverages variation in housing values within labor markets. I am able to implement this research design because there was a considerable amount of variation in house value changes within labor markets. Figure 1 displays ZIP code level changes in house values during the Great Recession in four different metropolitan areas: Atlanta, GA, Los Angeles, CA, Boston, MA, and Fresno, CA. Each map bins ZIP codes into quintiles based on their house value changes between January 2007 and December 2009. The gap between the 20th and 80th percentile in these metropolitan areas varies from 11.7% to 24.1%. There is a substantial amount of variation within each of the metropolitan areas, despite the fact that overall house value changes relative to the national distribution of house value changes were different between these metropolitan areas. Los Angeles and Fresno were in the top (4th) quartile of metropolitan areas by the size of house value shocks during the Great Recession, while Atlanta was in the 3rd quartile and Boston was in the 2nd quartile.

Exploiting within metropolitan area variation in house values allows me to control for

[5]Calculated using data from the Zillow Home Value Index (ZHVI). `https://www.zillow.com/research/data/`

local labor market conditions. Furthermore, within labor market variation in house values provides a cleaner experiment than using between labor market variation. Trends in house value growth in the period before the Great Recession are more similar within labor markets than between labor markets. Then trends were also far more similar during the post-recession recovery within labor markets. Within labor markets, the largest break in house value growth trends between ZIP codes experiencing high or low recession shocks occurs during the recession itself. This is not the case when evaluating the difference *between* high- and low-shock metropolitan areas. The choice to use variation in house value changes within labor markets instead of between labor markets ensures more stability in house price growth trends before and after the treatment used in my empirical specification. This is crucial since changes to parents' house values during these periods could also be relevant to the college and labor market outcomes of their children. Figure 2 illustrates the difference in house value growth trends within and between labor markets.

The left graph of Figure 2 compares house value growth trends of ZIP codes that experienced above median house value shocks within metropolitan areas to house value growth trends of ZIP codes that experienced below median shocks within metropolitan areas. Trends in house value growth prior to the recession were largely similar across ZIP code types. House value growth trends during the post-recession recovery have also been similar for ZIP codes that experienced relatively large or small shocks within metropolitan areas during the recession. The largest break in house value growth between the two types of ZIP codes occurs during the recession, and the house value index for ZIP codes experiencing large shocks during the recession has been permanently lower than the index for ZIP codes that had

smaller changes to house values. In contrast, the graph on the right shows that metropolitan areas that suffered large house value declines during the recession had far sharper increases in house values prior to the recession. These metropolitan areas have also experienced a stronger recovery in house values after the recession. The house value index for high shock metropolitan areas always remains greater than low shock metropolitan areas, despite the fact that they experienced more negative shocks during the recession. Figure 2 illustrates that exploiting within labor market variation in house values provides a cleaner experiment to study the impact of parents' house values on the adult outcomes of their children.

3 Data, Variable Construction, and Summary Statistics

3.1 Data and Sample Restrictions

I create a novel dataset by linking the 2005-2021 American Community Survey (ACS), 2000 Decennial Census Short Form, and the 2007-2010 Master Address File Auxiliary Reference File (MAF-ARF). My initial sample frame is adult respondents in the ACS aged 22-27 who were born between 1982 and 1994. These are individuals who turn 18 years old between 2000 and 2012. The lower age bound of 22 is chosen because it is the youngest age at which individuals (generally) began attaining bachelor's degrees, and I chose the upper age bound of 27 because it is the oldest age at which I observe members of all cohorts. Younger cohorts in my data were adolescents when the recession occurred. On the other hand, older cohorts

were old enough to have completed their initial bachelor degree programs when the recession began, and will serve as the control group when estimating the generalized difference-in-differences specification.

I use Protected Identification Keys (PIKs) that identify individuals across census administrative records and survey data sets to link individuals in my initial sample frame to themselves in the 2000 Decennial Census.[6] The match between the ACS and the 2000 Decennial Census creates intergenerational linkages between the initial sample frame and their parents since members of the initial sample are between the ages of 5-18 when their families are surveyed in the 2000 Decennial Census. I keep only linked individuals who are biological, adoptive, or step-children of the surveyed householder in the Decennial Census. I further restrict my sample to children of parents who are homeowners in the 2000 Decennial Census. I limit my sample to the children of homeowners because my natural experiment uses changes to house values to study the influence of parents' resources on children.

At this point, the data includes outcomes during young adulthood, a linkage between parents and children, and parents' ownership status. The next step determines a household's geographic location during the Great Recession. I could use the household's location in the 2000 Decennial to proxy for its location during the recession, but this would result in measurement error of the housing wealth shock experienced by families if they move between the Decennial Census and the Great Recession. This measurement error could lead to my results suffering from attenuation bias (see Stuart (2022)). To overcome this challenge and

[6]PIKs are assigned using a probabilistic match based on name, date of birth, address, and sex. I exclude individuals from my sample if they do not have a unique PIK in an ACS survey year, or if they are linked to a PIK in the 2000 Decennial that is non-unique and does not have a unique assigned high score from the Census Bureau's Person Identification Validation System (PVS).

obtain more precise data on the geographic location of households in my data during the Great Recession, I link my sample to the MAF-ARF. The Master Address File (MAF) is a Census Bureau dataset that provides information on all known living quarters in the United States. It includes Census Bureau address identifiers (MAFIDs) and geographic location codes. The MAF Auxiliary Reference File (MAF-ARF) assigns a MAFID to all individuals who are associated with a domicile in a given year (Sullivan and Genadek, 2022).[7]. A description of the procedure I use to link my analysis sample to the MAF-ARF can be found in Appendix Section A.1.

I use the location from the MAF-ARF linking procedure to assign a childhood geographic location to cohorts turning 18 between 2007 and 2012. These cohorts were 18 or younger when the recession began in 2007, and hence their initial college investment decisions were influenced by parents' house value shocks during the recession. Using the geographic location from the MAF-ARF instead of the 2000 Decennial Census removes a significant amount of measurement error in geographic location; 23.3% of the treatment group sample moved ZIP codes between 2000 and 2007. Earlier cohorts (those turning 18 between 2000 and 2006) are assigned geographic locations based on their locations in the 2000 Decennial Census. Respondents from these cohorts were 18 or older when the recession began, and made their initial college-going decisions prior to the beginning of the recession. I am not explicitly interested in the impact that parents' house value shocks during the recession had on these cohorts, instead these cohorts will serve as control groups in my research design. I am simply

[7]The MAF-ARF is created by finding PIK and MAFID combinations from several sources of federal administrative data, including tax returns, as well as Census Bureau survey data. PIKs are not uniquely assigned to MAFIDs; the mean MAFIDs per PIK was 1.497 in 2007 (Sullivan and Genadek, 2022)

interested in their location during adolescence.

The final step of my sample construction merges the census data with external data from two sources. I use data from the National Historical Geographic Information System (NHGIS) to control for the economic and demographic characteristics of childhood Census tracts and ZIP codes. I also include data from the Zillow Home Value Index (ZHVI). I use the ZHVI to construct my measures of local house value changes. The two ZHVI data series that I utilize in my analysis are the "All Homes (SFR, Condo/Co-op) Time Series, Smoothed, Seasonally Adjusted" series for both ZIP codes and metropolitan areas. Both of these data series report typical values for homes in the 35th to 65th percentile range for each month. The ZHVI is constructed using Zillow's home value estimates (Zestimates) within the given geographic unit. These estimates are created using property level data such as listing information, tax assessments, and sales histories, along with home characteristics such as size and location. The advantage of using Zillow's ZHVI is that it should improve small area value signals when compared to traditional House Price Indexes (HPIs), which require multiple transactions at the same property to construct an index. The ZIP codes with ZHVI data during the Great Recession covered 81% of the U.S. population. My sample only includes individuals who are assigned childhood geographies that I am able to match to the NHGIS and Zillow home value data.

My baseline sample is further restricted to individuals born in the intervals 1983-1984 or 1989-1994. These are individuals who turn 18 in 2001-2002 or 2007-2012. The 2001-2002 age-18 cohorts will serve as a control group, while the 2007-2012 age-18 cohorts are treated by parental house value shocks during the Great Recession. There are approximately 1,730,850

respondents in the public use 2005-2021 ACS who were born in these cohorts and aged 22-27; my baseline sample includes roughly 937,000 respondents. This represents about 54% of the initial potential sample frame. Additional cohorts are utilized later to form alternative samples to confirm that my results are robust to alternative control cohort definitions.

3.2 Variable Construction

All outcome variables are constructed using information from the ACS. I will begin by describing the construction of indicator variables related to undergraduate college education. Bachelor's Degree Attainment is an indicator that is equal to 1 if the respondent's highest level of education is a bachelor's degree or greater. Associate's Degree Attainment is an indicator that is equal to 1 if the highest level of education is an associate's degree. Current Undergraduate Enrollment is an indicator that is equal to 1 if an individual is currently enrolled in a two or four year degree undergraduate program[8]. Any College Education is an indicator that is equal to 1 if an individual's highest level of education is "some college" or greater. Degree Non-completion is an indicator that is equal to 1 if an individual's highest level of education is some college but not a college degree and they have not been enrolled in college program for the last three months. B.A. attainment, degree non-completion, and A.A. attainment are mutually exclusive outcomes, but the attendance outcome is not mutually exclusive to the degree attainment outcomes. It is possible for someone with either an A.A. degree or even a B.A. degree to be currently enrolled in an undergraduate program.

I also study a set of outcomes related to the labor market, migration, and graduate

[8]The ACS does not differentiate between two and four year degree programs when asking about current undergraduate college enrollment.

school experiences of young adults. I use the adjusted personal earnings variable to measure Earnings. This variable is inflation adjusted to 2011 dollars. Full-Time Employment is an indicator equal to 1 if the respondent was employed full time in the previous year. Current Graduate School Enrollment is an indicator that is equal to 1 if currently enrolled in a graduate school program. Any Graduate School Attendance is an indicator that is equal to 1 if the respondent is currently enrolled in graduate school or whose highest education level is a master's degree or greater. I am unable to use the ACS to identify if an individual has dropped out of a graduate school program. Parental Co-residence is an indicator equal to 1 if the respondent is the child of a householder in the ACS. I consider this a labor market outcome given the results of Kaplan (2012), who finds that young adult children use living at home as informal insurance against labor market risk. Out-of-State Migration is an indicator equal to 1 if the individual does not live with their parents and they live in a different metropolitan area (defined by Core-Based Statistical Areas) that is in a different state when surveyed in the ACS than they did as a child.

I use the ACS, Decennial Census, and the MAF-ARF to construct a set of controls that I include in my specifications. The controls constructed using my census sample are standard and straightforward, and more details about them can be found in Appendix Section A.2. The one variable that I explain here is the variable titled Move; this is an indicator equal to 1 if the respondents' family moved between the 2000 Decennial Census the and when they are observed in the MAF-ARF. Specifically, this value will be equal to 1 if I match them to a different ZIP code in the MAF-ARF then their ZIP code in the 2000 Decennial Census. Note that this variable will only be equal to 1 for individuals that I observe alongside their

parents in the MAF-ARF.

I use data from NHGIS to construct controls for the economic and demographic conditions of an individual's childhood Census tract and ZIP code. NHGIS uses the Decennial Censuses and the ACS to construct measures of population characteristics within geographic units. I use year 2000 data derived from the 2000 Decennial Census to measure Census tract characteristics. I also create measures for how the following ZIP code characteristics changed between 2000 and the years 2007-2011 pooled together: share of black residents, share of white residents, share of age 25+ residents with a bachelor's degree, share of age 25+ residents with a high school diploma, poverty rate, unemployment rate, and log median household income. The choice to measure the change between 2000 and 2007-2011 is a data restriction. All NHGIS economic and education variables after 2000 are constructed using the ACS, and ZIP codes are geographic units that require 5 years of ACS data to be pooled together to create geographic measures. These measures capture how ZIP code characteristics changed prior to and during the Great Recession.

I use Zillow's Home Value Index (ZHVI) to construct the following variable to measure local house value changes within a given geographic unit:

$$H_g^{07-09} = 10 * (\frac{ZHVI_g^{Dec2009} - ZHVI_g^{Jan2007}}{ZHVI_g^{Jan2007}}), \tag{2}$$

where $ZHVI_g^{Jan2007}$, $ZHVI_g^{Dec2009}$ are ZHVI values (measured in \$s) in geographic unit g during January 2007 and December 2009, respectively. This geographic unit is either a metropolitan area (m) or 5-digit ZIP code (z). Therefore, H_g^{07-09} measures the 10% change

in typical house values in g between January 2007 and December 2009 (i.e. $H_g^{07-09} = 1$ means that house values in g grew by 10% between 2007 and 2009). The choice to use the time period of 2007-2009 to define the housing value change is consistent with the prior literature studying the Great Recession (e.g. Yagan (2019); Rinz (2022)), which uses local economic shocks measured between 2007 and 2009 as the key independent variable in their empirical specifications. H_g^{07-09} is a proxy for the housing value shock experienced by households during the Great Recession.

3.3 Summary Statistics

Table 1 presents summary statistics on the outcomes of the main analysis sample. There are roughly 937,000 individual-level observations in the sample. Table 1 reports the averages for each of the outcomes studied across three age bins; 22-23, 24-25, and 26-27. There is considerable growth in bachelor's degree attainment, earnings, and employment with age. There is also considerable growth in out-of-state migration and any graduate school experience. A.A. attainment, degree non-completion, and current graduate school attendance remain fairly flat over the age profile, while undergraduate enrollment and parental co-residence decrease sharply as young adults get older. Appendix Table B.1 reports summary statistics for selected control variables, and reports summary statistics by treatment or control cohort status. The mean ZIP code house value change between January 2007 and December 2009 experienced by households in the main analysis sample is -15.81%, and the mean metropolitan area house value change for households in the main analysis sample -15.84%. The 10th percentile ZIP code change is -40.02%, and the 90th percentile change is 1.06%. The 10th

percentile metropolitan area change is -40.31%, and the 90th percentile change is -0.49%.

4 Empirical Design

I estimate the impact of parents' house value shocks during the Great Recession on post-secondary human capital investment and labor market outcomes using a generalized difference-in-differences approach. The first difference compares individuals whose parents lived in ZIP codes experiencing larger or smaller declines in house values, and the second difference compares individuals who were older or younger during the Great Recession. The specification is given by:

$$y_{it} = \beta_j[age_{j(i)} \times treat_{c(i)} \times H_{z(i)}^{07-09}] + x_{i,z(i),tr(i)}\alpha + \theta_{c(i),m(z(i))} + \delta_t + \gamma_{z(i)} + \epsilon_{it}. \quad (3)$$

The dependent variable, y_{it}, is a college education or labor market outcome for individual i who appears in the ACS in survey year t. The primary outcomes I study are bachelor's degree attainment, earnings, and full-time employment. I also study associate's degree attainment, degree non-completion, any college attendance, current undergraduate attendance, current graduate school attendance, any graduate school attendance ever, parental co-residence, and out-of-state migration.

The key independent variable $H_{z(i)}^{07-09}$ measures the 10% percent change in ZIP code z's housing values during the Great Recession and is defined in Equation (2). This variable is interacted with a vector of age bin indicators $age_{j(i)}$ and the treatment group indicator

$treat_{c(i)}$. I include three age bin indicators in Equation (3) indexed by j; i.e. indicators equal to 1 if i is aged 22-23, 24-25, or 26-27 when their outcomes are observed. The purpose of these age indicators is to study if the treatment effect of parents' house value shocks is persistent and/or varies during young adulthood. Treatment group status is determined by i's cohort c, which I index by the year that i turns 18. $treat_{c(i)}$ is equal to 1 if i's cohort $c \in [2007, 2012]$, while the control group includes individuals that turn 18 in the years 2001 or 2002[9]. Respondents in the treatment group make their initial post-secondary investment decisions during or in the immediate aftermath of their parents experiencing resource shocks from the Great Recession. In contrast, the control group is (at least) 22 years old at the beginning of the recession, and have made their initial college investment decisions prior to the recession. Their initial college investment choices are untreated by parents' house value shocks that occurred during the Great Recession. It is also important to note that the cohorts included in the control group made their initial college entry decisions prior to the height of the housing bubble. It is problematic to use later cohorts in the control group because these cohorts are treated by the housing bubble, which is correlated with $H_{z(i)}^{07-09}$. Later cohorts could be treated by the recession itself, making them an unsuitable control group. Nonetheless, I will show that my results are robust to alternative control group definitions.

The parameter of interest is β_j, which varies by age. β_j measures the impact of a 10%

[9]I opt for a control group comprised of children of homeowners from earlier cohorts rather than children of non-homeowners to avoid measurement error. I am only able to observe homeownership status in the 2000 Decennial Census, and am unable to observe it during the recession years of 2007-2009. Year 2000 homeownership status is a strong proxy for homeownership in 2007 for parents who were homeowners in 2000, but is a poor proxy for those who were not homeowners in 2000. Using the public-access PSID, I calculate that 91% of homeowners in the 2001 survey with children born in the years 1989-1994 are still homeowners in 2007, where as only 63% of parents who are renters in 2001 are still renters in 2007.

change in ZIP code z's house values during the Great Recession on college and labor market outcomes at ages j. The fixed effects and controls in Equation (3) that control for labor market influences on y_{it} allow me to interpret the estimands β_j as the influence of changes to parents' house values on a given outcome. I allow this parameter to vary by age in order to confirm that my results are persistent and are not driven by parents' resources facilitating earlier degree completion or initial higher earnings. When the outcome is bachelor's degree attainment, earnings, or full-time employment, I expect that β_j will be positive at later ages. This would imply that a more positive (or less negative) change to parents' housing values during the Great Recession increases degree attainment and future labor market outcomes.

I include survey-year fixed effects δ_t and ZIP code fixed effects $\gamma_{z(i)}$ in Equation (3). δ_t controls for calendar-year trends in outcomes, which is particularly relevant when y_{it} is a labor market outcome. $\gamma_{z(i)}$ controls for time-invariant ZIP code characteristics that are related to both the outcome and the local house value shock that i's household experienced. These characteristics can include the average quality of schools in z, as well as the economic circumstances and demographic characteristics of the residents of z. Controlling for ZIP code level differences in outcomes is crucial in light of the results of Mian and Sufi (2009), who find that mortgage default during the U.S. housing crisis was concentrated in ZIP codes with high ex-ante rates of subprime borrowers.

A key feature of my empirical model is the inclusion of metropolitan area-by-cohort fixed effects $\theta_{c(i),m(z(i))}$. The metropolitan area-by-cohort fixed effects allow for cohort trends in y_{it} to vary flexibly across metropolitan areas. Crucially, these fixed effects control for local labor market conditions that influence both local house price changes as well as i's

college investment decisions and labor market outcomes. The inclusion of the fixed effects $\theta_{c(i),m(z(i))}$ and $\gamma_{z(i)}$ makes Equation (3) a difference-in-differences specification, and the fact that the cohort effect varies across metropolitan areas means that I am exploiting *within* labor market variation in house value changes. These fixed effects also control for national and state level changes in financial aid policies or tuition pricing that could be related to house value changes.[10]

The vector $x_{i,z(i),tr(i)}$ includes indicators for race/ethnicity[11], sex, parent(s)' marital status in the year-2000, sex of the parent householder, type of child (biological or adopted, with step-child serving as the omitted category), and an indicator equal to 1 if the household moved to a different ZIP code between the 2000 Decennial Census and the year 2007 when they are observed in the MAF-ARF. $x_{i,z(i),tr(i)}$ also includes a set of controls for the following characteristics of i's year 2000 Census tract tr: log median household income, poverty rate, share of black population, share of white population, age 25+ bachelor's degree and high school diploma attainment rate, and unemployment rate. These individual, household, and Census tract variables are included since they could be correlated with both the outcomes of interest and the house value shock experienced by i's parents. I also include an age quadratic to control for life-cycle trends in outcomes y_{it}.[12]

A remaining identification concern is that within metropolitan area house value changes

[10]Prominently, federal student loan borrowing limits were raised in 2007 and 2008. Furthermore, Barr and Turner (2013) show that tuition prices at flagship state institutions are positively related to unemployment rate changes (i.e. states with larger unemployment rate increases had larger increases in tuition prices).

[11]Dummies for Asian, Black, Hispanic, white, with all other races/ethnicity combinations serving as the omitted category.

[12]The fact that cohorts are indexed by the year they turn 18 means that I observe members of a given cohort c at two ages during a calendar year t. Thus, the age quadratic is not collinear with the cohort and time fixed effects.

are reflective of within labor market differences in recession shock intensity or demographic changes. The ZIP code fixed effects $\gamma_{z(i)}$ control baseline differences across ZIP codes within metropolitan areas, but estimates would be biased if changes to ZIP code characteristics prior to or during the Great Recession are correlated with house value changes. For example, if a relatively high house value shock ZIP code within a metropolitan area also suffers a relatively high within metropolitan area shock to household income, then it is unclear if the estimated impact of house value changes is driven by the decrease in the value of the parents' housing asset or decreases in other household resources such as income. To address this concern, $x_{i,z(i),tr(i)}$ includes a set of controls for changes in the characteristics of i's ZIP code z before and during the Great Recession. I include the change in $z(i)$'s share of black and white residents, age 25+ bachelor's degree and high school diploma share, poverty rate, unemployment rate, and log household income between 2000 and data pooled from 2007-2011. The ZIP code controls are time invariant, and therefore are interacted with $treat_{c(i)}$ to facilitate the inclusion of ZIP code fixed effects $\gamma_{z(i)}$.

A key identification assumption is conditional parallel trends. In other words, conditional on the controls included in Equation (3), the average difference in outcomes between ZIP codes would have remained the same if they received the same house value shock $H_{z(i)}$. I will reiterate how a pair of features of Equation (3) support the assumption of conditional parallel trends and allow me to interpret the estimands of β_j as the causal impact of parents' house values. Metropolitan area-by-cohort fixed effects control for the impact that local labor markets have on y_{it}, which means that leveraged variation in housing prices is not driven by the recession's general labor market scarring effect or its impact on the opportunity costs

of education investment. The next concern was that within labor market variation in house values was related to endogenous factors that drove differential house value changes during the recession within metropolitan areas. Augmenting the ZIP code fixed effects with the set of ZIP code variables that control for within labor market economic conditions plausibly makes the within metropolitan area house value variation like-random. These two key features, along with the additional controls and fixed effects, result in a model that isolates the effect that the recession had on outcomes via its impact on parents' housing wealth from other factors such as local labor market conditions. Standard errors are clustered by ZIP code to account for serial correlation within ZIP codes.

5 Results

5.1 Impact of Parents' House Values on Bachelor's Degree Attainment

The first key finding of this paper is that parents' house values increase bachelor's degree attainment. Figure 3 reports the coefficient estimates of β_j of Equation (3) when the outcome is bachelor's degree attainment. These estimates represent the estimated impact of parents' house values on bachelor's degree attainment. I find that the impact of parents' house values on bachelor's degree attainment varies over the studied age profile of 22-27. The impact of a 10% increase in house values is slightly negative and not statistically significant at ages 22-23, but it increases bachelor's degree attainment by 0.571 percentage points (p.p.) at

ages 24-25 and by 0.399 p.p. at ages 26-27. These estimates are statistically significant at a 95% level of confidence, and translate into a 1.24% and 0.80% increase in bachelor's degree attainment relative to mean attainment rates in those age bins respectively. The fact that results are positive and statistically significant at the later ages studied means that parents' house values lead to a persistent increase in human capital accumulation.

Alternative Specification with Fewer Parametric ZIP Characteristics

The estimates from the baseline specification are interpreted as the impact of parents' house value changes during adolescence on bachelor's degree attainment. Focusing on parents' house values alone understates the total impact that parents' resource shocks during the recession had on the outcomes of their children. Shocks to other household resources, such as reduction in income from job loss, are almost certain to be salient as well. To get a better idea of the broader impact of recession shocks experienced by households, I estimate two alternative specifications of Equation (3) that drop some or all of the parametric ZIP code controls. The first alternate specification includes only parametric ZIP code controls for changes in the white and black share of residents and changes in shares of residents with H.S. diplomas and B.A. degrees between 2000 and 2007-2011. The second alternate specification does not include any parametric ZIP code level controls. These alternate specifications are otherwise identical to the baseline empirical specification.

Estimates of β_j from the alternate specifications will be biased in the sense that they no longer isolate the impact of parents' house values on B.A. attainment. This bias arises from the fact that house value changes during the recession can be correlated with recession

intensity within metropolitan areas. However, it is likely that the estimates of β_j are being biased upwards by changes in ZIP code characteristics that impact the B.A. attainment of children via the impact of these changes on parents.[13] The metropolitan area-by-cohort fixed effects remain, which means that alternative specifications still control for the impact of local labor markets on college attainment. Furthermore, these alternative specifications still include ZIP code fixed effects, which means baseline initial differences across ZIP codes are still controlled for. The estimates of β_j from the alternative specifications now reflect the impact of housing values and other changes to parents' financial resources, such as employment or income. The estimates could also be influenced by changes to neighborhood or schools, but these changes should not have a large impact on adult outcomes (see Footnote[13]).

Figure 4 reports the results from the three different specifications of Equation (3) that vary by the parametric ZIP code controls included during estimation. The estimates from all three specifications follow the same pattern: the estimated impact of parents' housing wealth on bachelor's degree attainment is statistically indistinguishable from 0 at ages 22-23, but is positive at ages 24-25 and 26-27. As expected, the less restrictive the model, the less

[13]In addition to influencing the wealth of parents, it is possible that changes to house values also reflect and/or impact the demographic characteristics and quality of high schools that adolescents are exposed to. The baseline specification parametrically controls for demographic shifts by including controls for changes to race and education shares, but the least conservative alternative specification does not. However, it is unlikely that changes to school quality or neighborhood demographics that are correlated with house value changes during the recession would have a large effect on bachelor's degree attainment in the analysis sample that I study, for several reasons. First, a portion of the treatment group has already left high school or are very late in the their high school careers during the Great Recession, and thus are untreated by impacts that the recession has on their schools or neighborhoods. Secondly, there is mixed evidence on the impact of neighborhoods on the impact of neighborhoods on the adult outcomes of adolescents (Chetty et al., 2016; Chyn, 2018). Finally, while it is possible that house value changes during the recession impact school quality, the effect sizes are small relative to the observed house value changes. Gilraine et al. (2023) estimates that a 100% increase in home values in a 5-year period increases school value added by 0.16 standard deviations. Extrapolating from Gilraine et al. (2023) and Chetty et al. (2014a), a 10% increase in home values over a two year period would imply an 0.033 p.p. increase in college attendance, and an increase in earnings of 0.052%. These are small effect sizes.

conservative the estimate is in all 3 age bins. When including demographic controls only, a 10% increase in ZIP code house values increases bachelor's degree attainment by 0.680 p.p. (1.49%) at ages 24-25, and by 0.507 p.p. (1.02%) at ages 26-27. When no ZIP code controls are included, a 10% increase in ZIP code house values increases bachelor's degree attainment by 0.785 p.p. (1.72%) at ages 24-25 and by 0.613 (1.23%) at ages 26-27. The least restrictive estimate at age 26-27 is 53.75% larger than the baseline result.

Contextualizing Magnitude of Estimates

The magnitude of the estimated impact of changes to parents' house values on bachelor's degree attainment is large relative to what one might reasonably expect when compared to studies of other interventions impacting college attainment. These studies include changes to parents' financial resources or changes to financial aid policies that impact borrowing limits or the price of college. For the most part, these papers focus on samples comprised of individuals that are close to the college attainment decision margin. These include samples comprised only of college-goers (Lovenheim and Reynolds, 2013; Hotz et al., 2023), credit-constrained college-goers (Black et al., 2020), or those who apply for college financial aid (Bettinger et al., 2019). In contrast, my sample is drawn from the population of children of homeowners. Large portions of my sample are not close to the margin of degree attainment, and thus will experience no treatment effect when their parents' resources change. Therefore, to contextualize my results, I will compare them to two papers that study degree attainment using samples drawn from broad populations.

The first paper that provides comparable results in a distinct but related context is

Stuart (2022). This paper finds that a 10% decrease in county-level earnings per capita reduces long-run bachelor's degree attainment by 2.7 p.p. for children who were between the ages of 11-19 during the 1980-1982 double-dip recession. On the surface, this effect appears to be substantially larger than the findings in my paper. However, a 10% change in earnings per capita is a large treatment dosage relative to typical changes in county-level earnings per capita during the 1980s recession. The average per capita income change between 1980-1982 in counties suffering above-median recession shocks during was -8.9%, slightly smaller than the treatment dosage of a -10% change. By comparison, a 10% decrease in home values during the Great Recession is a small treatment dosage relative to typical changes to housing values during the recession. The average house value change between January 2007 and December 2009 in ZIP codes that suffered above-median house value shocks was -30.7%. Using this value as the treatment dosage, my results imply that house value shocks alone during the Great Recession decreased bachelor's degree attainment by 1.2 p.p. when using the most restrictive estimate from Equation (3). The estimate that is most comparable to Stuart (2022) is from the least restrictive version of Equation (3), since this estimate represents a more complete estimate of the recession's local economic shock. When using this estimate, a 30.7% decrease to local housing values reduce B.A. attainment by 1.88 p.p. Stuart (2022)'s estimate is within the confidence interval of my re-scaled estimate.

It is somewhat surprising that the least conservative estimate in my paper is statistically indistinguishable from the result in Stuart (2022). The results in Stuart (2022) are driven by the entire impact of the recession before adolescence, when parents' resources and local environments have the largest impact on human capital development. In contrast, my results

reflect the impact of parents' resources during adolescence, and are likely driven by the impact of changes to resources on college financing. In other words, the treatment studied in my paper is arguably "weaker" and impacts an older population that one would expect to be less sensitive to treatment.

Another useful comparison comes from Dynarski (2008), which studies the impact of large tuition subsidy programs from Arkansas and Georgia during the 1990s on population-level college degree attainment. This paper is particularly relevant since it studies the impact of college financing on degree attainment. The author finds that these programs increased degree attainment by about 3 p.p. My estimates are 40%-62.67% of this treatment effect for households that experience the typical large shock to housing wealth. My estimates are substantial in comparison considering that the policies studied by Dynarski (2008) are direct tuition payments that cover nearly all or the entire tuition costs faced by most households. In contrast, a household's marginal propensity to spend housing wealth on their child's education is far lower than 1. Hotz et al. (2023) estimate (imprecisely) that a $10,000 increase in home equity increases parents' transfers to their children's college education by $135. The average house value change for ZIP codes experiencing above-median shocks was about $100,000. Extrapolating from Hotz et al. (2023), households suffering a "typically severe" housing wealth shock decrease spending on college by $1,350, and my estimates imply that children from these households experience a decrease in bachelor's degree attainment of 1.2 p.p. This is a large estimate given that the average nominal cost of tuition and fees at a 4-year college in 2007-2008 was about $11,500[14].

[14] https://nces.ed.gov/programs/digest/d22/tables/dt22_330.10.asp

5.2 Alternative Approach Using Metropolitan Area Variation

Another finding of this paper is that it is important to exploit within metropolitan area variation in house values in order to control for the influence of local labor markets. I compare the estimates from the baseline specification to a specification that uses *between* metropolitan area variation in house values. This specification is similar (but not identical)[15] to approaches used in the prior literature studying the influence of parents' housing wealth on post-secondary investment and labor market outcomes. The generalized difference-in-differences equation that I estimate is given by:

$$y_{it} = \lambda_j[age_{j(i)} \times treat_{c(i)} \times H_{m(i)}^{07-09}] + x_{i,tr(i)}\psi + \sigma_{c(i),s(m(i))} + \tau_t + \eta_{m(i)} + u_{it}. \quad (4)$$

I study the same set of outcomes y_{it} as I do in Equation (3). The treatment indicator $treat_{c(i)}$ and the vector of age indicators $age_{j(i)}$ are defined as they are in Equation (3), and survey-year fixed effects τ_t are included as well. The vector $x_{i,tr(i)}$ no longer includes ZIP code controls but otherwise is defined as it is in Equation (3). The key independent

[15]Previous papers use the PSID or the NLSY, and as a result, have richer information on household socioeconomic circumstances and the exact wealth shock experienced by households. In contrast, I am limited to the household information that is reported in the 2000 Decennial Short Form. The benefit of my data is that it produces samples that are far larger than those used in the prior literature (e.g. up to 1,500Xs larger when compared to the baseline sample in Cooper and Luengo-Prado (2015)). This increases the precision of my estimates of Equation (4). Furthermore, the state-by-cohort fixed effects I include in Equation (4) non-parametrically control for local labor market conditions in a more general way than the parametric state-level controls used in the prior literature. It should be noted that the prior literature does use specifications that include parametric controls for economic conditions at levels of geography that are finer than the state level (Lovenheim and Reynolds, 2013; Cooper and Luengo-Prado, 2015; Hotz et al., 2023).

variable, $H_{m(i)}^{07-09}$, measures the house value change in metropolitan area m between January 2007 and December 2009, and is defined by Equation 2. Metropolitan fixed effects $\eta_{m(i)}$ are included to control for time-invariant metropolitan area characteristics that are correlated with both y_{it} and $H_{m(i)}^{07-09}$. Finally, state-by–cohort fixed effects $\sigma_{c(i),s(m(i))}$ flexibly control cohort trends in y_{it} across states. The inclusion of $\sigma_{c(i),s(m(i))}$ and $\eta_{m(i)}$ make Equation (4) a difference-in-differences specification that exploits within state variation in house values.

The parameter of interest is λ_j. This parameter measures the impact of a 10% change in metropolitan area housing values on the outcome variable. Unlike Equation (3), these estimands cannot be interpreted as the influence of parents' housing wealth on labor market or post-secondary outcomes. In particular:

$$\mathbb{E}[u_{it}|age_{j(i)} \times treat_{c(i)} \times H_{m(i)}^{07-09}, x_{i,tr(i)}, \sigma_{c(i),s(m(i))}, \tau_t, \eta_{m(i)}] \neq 0 \tag{5}$$

A major concern is that the estimates of λ_j will suffer from omitted variable bias due to the fact that metropolitan area house values are related to local economic conditions. The fixed effects and time-varying controls included in Equation (3) do not sufficiently control for the confounding influence of the local economy. The primary concern is that state-by-cohort fixed effects are not sufficient to control for local economic conditions, since the relevant local labor market is a geographic unit that is smaller than a state. As discussed in Section 1, the direction of bias could be positive or negative when the outcome is B.A. attainment. In the context of the Great Recession, the correlation between house values and the opportunity cost of college education is a major concern which would result in a downward bias. Standard

errors are clustered by metropolitan area to allow for serial and spatial correlation within metropolitan areas.

Results: Alternative Specification Using Between Labor Market Variation in House Values

Figure 5 reports the coefficient estimates of λ_j of Equation (4) and compares them to the baseline estimates of β_j from the most restrictive version of Equation (3). At all ages, the coefficient estimates of λ_j from Equation (4) are closer to 0 than the corresponding β_j estimates from the baseline specification. The estimate of λ_{26-27} is statistically indistinguishable from 0 when estimating Equation (4), which suggests that the impact of local housing values on bachelor's degree attainment is not persistent. These results provide suggestive evidence that estimates from a specification that does not control for local labor market conditions using metropolitan area-by-cohort fixed effects are biased downwards.

A plausible explanation for why previous papers have uncovered statistically significant estimates without using within labor market variation in house values is that many of these papers focus on cohorts that made their college decisions during the 2000s housing boom. This is a relevant distinction because there was a weaker relationship between local economic conditions and house values during the 2000s housing boom than there was in the Great Recession. Figure 6 reports results from simple regressions of average metropolitan area unemployment changes on average metropolitan area percent changes in house values during the 2007-2009 recession and the 2000-2006 housing boom. A steeper slope indicates a stronger relationship between unemployment and house values, while a flatter slope indicates a weaker

relationship.

As expected, the slope of the simple regression coefficient in both time periods is negative, implying that there is a negative relationship between local unemployment and housing markets (and hence, a positive relationship between local house values and the health of the local labor market). Figure 6 also shows that labor markets with similar changes to unemployment rates experienced vastly different house value growth during the housing boom, while areas experiencing large house value declines during the Great Recession experienced larger local unemployment shocks. The estimated relationship between local house values and local unemployment changes was nearly 5 times larger during the recession than it was in housing boom. The fact that the association between local labor markets and house values was weaker during time periods previously studied means that it was less costly to not use within metropolitan area variation in home values to control for the influence of local labor markets. Consistent with this logic, Charles et al. (2018) find that metropolitan area housing demand growth during the 2000s housing boom did not impact bachelor's degree enrollment or attainment.

5.3 Robustness of B.A. Attainment Results to Alternative Control Groups

I perform two robustness checks to confirm that my results are robust to alternative control cohort definitions. I estimate Equation (3) and Equation (4) using two samples that are slightly different. The first robustness check uses cohorts turning 18 in 2000-2002 as the

control group and and the second check uses cohorts turning 18 in 2001-2003 as the control group. The first robustness check adds one earlier cohort, and the second robustness check adds one later cohort to the baseline sample. Figure 7 reports results from these robustness checks for both the ZIP code and metro-area specifications.

Qualitatively, the estimated impact of parents' house values on B.A. attainment is similar across both specifications regardless of the control cohorts used. The coefficient estimates at ages 22-23 are negative in all ZIP and metro-area specifications, and are generally statistically indistinguishable from 0. The exceptions are the ZIP specification using cohorts turning 18 in 2000-2002 and the metro-area specification using cohorts turning 18 in 2001-2003 as control groups. The age 22-23 coefficient estimates in these specifications are statistically significant at the 10% level.

All age 24-25 estimates using the ZIP code specification are statistically significant at the 5% level, and all age 26-27 estimates are at least statistically significant at the 10% level. All age 24-25 and age 26-27 coefficient estimates from the metropolitan area specification are smaller than their counterparts from the main ZIP code specification, and none of the age 26-27 estimates are close to statistically significant. Taken together, these results provide robust evidence that parents' home values during the Great Recession increased bachelor's degree attainment. They also strengthen the suggestive evidence that results from the approach that does not exploit within labor market variation in home values yields biased estimates.

I also check to see if my results are robust to the inclusion of the 2003 and 2004 age-18 cohorts to the main analysis sample. For the most part, results are robust. However the age 26-27 coefficient estimate is only statistically significant in the ZIP code specification

when some or all of the parametric ZIP code controls are dropped. It is not a surprise that the results are less robust when including later cohorts in the control group, as these cohorts are treated by both the 2000s housing boom and the Great Recession. The fact that there are cohorts included in the control group that are also treated by the Great Recession means the expected direction of bias when the outcome is bachelor's degree attainment is negative. A more thorough discussion and the results of this robustness exercise can be found in Appendix Section A.4.

5.4 Impact of Parents' House Values on Other College Outcomes

Any College Attendance

Table 2 reports estimates of the impact of parents' house values on any undergraduate college attendance, associate's degree attainment, degree non-completion, current undergraduate enrollment, graduate school enrollment, and any graduate school education. The impact of parents' housing wealth on bachelor's degree attainment is not driven by an increase in college attendance. In fact, adolescents whose parents suffered larger house value declines during the Great Recession were slightly *more likely* to have any college attendance. This result is not surprising in light of the findings of Barr and Turner (2013), who show that the largest increase in the share of enrollment during the recession was concentrated in low cost 2-year degree institutions.

Parents' house values also influence other margins of college investment. Parents' home values decrease the likelihood that a child drops out of college, initially substitutes into a

lower cost two-year degree, or is enrolled in an undergraduate education program in their mid-to-late 20s. Furthermore, parents' house values also decrease enrollment in graduate school programs.

Associate's Degree Attainment

A 10% increase in parents' house values decreases the likelihood that an associate's degree is an individual's highest degree by 0.19 p.p. (2.13%) at ages 22-23. This estimate is statistically significant at a 90% level of confidence (p-value = 0.07). The implication is that children whose parents' suffer large house value shocks during adolescence are more likely to initially substitute into lower cost two-year degree programs and away from higher cost four-year degree programs. This result is consistent with prior work which finds that parents' resources (Belley and Lochner, 2007) or increased financial aid (Bettinger et al., 2019; Black et al., 2020) increase attendance and attainment of 4-year degree programs and reduce attendance and attainment from 2-year degree programs. This pattern of substitution fades out over the studied age profile: coefficient estimates become smaller (in absolute terms) at ages 24-25 and 26-27, and are statistically indistinguishable from 0.

Degree Non-completion

Parents' house values during the Great Recession also substantially decrease the likelihood that a child leaves an undergraduate program without a degree across the entire age profile studied. A 10% increase in parents' house values during adolescence reduces the likelihood that a child has left an undergraduate program without a degree by 0.59 p.p. (3.73%)

at ages 26-27. The fact that parents' resources decrease college drop out is a sensible result. Conditional on college enrollment, parents' resources have been shown to increase degree completion (Lovenheim and Reynolds, 2013; Hotz et al., 2023). Investment in higher education exposes individuals to income risk due to the fact that many students finance tuition costs with student loans and have uncertain future wages. Parents accumulate financial resources to insure against their children's income risk (Boar, 2021). It is unsurprising that children are more likely to drop out of a college program when their parents suffer negative shocks to the value of their housing asset, and thus have fewer resources to help pay for their child's college or insure against the income risk of their children.

Current Undergraduate Enrollment

The final result related to undergraduate post-secondary education is that parents' home values decrease enrollment in undergraduate college programs in a child's mid-to-late 20s. The impact of parents' house values is small, positive, and statistically indistinguishable from 0 at ages 22-23, but negative at later ages. A 10% increase in parents' house values decreases undergraduate enrollment by 0.79 p.p. (5.35%) at ages 24-25 and by 0.49 p.p. (5.26%) at ages 26-27. The timing of this result is consistent with the impact of parents' house values on B.A. attainment. Children of parents suffering large and small house value shocks during the Great Recession are just as likely to be enrolled in an undergraduate program at ages 22-23. Children of parents with greater housing wealth are more likely to be completing their bachelor's degree programs at this time, and realize a persistent increase in B.A. attainment. Meanwhile, children of parents who suffered large negative house value shocks during the

recession are less likely to complete a B.A. program, and are more likely to be accumulating human capital via higher education in their late 20s or to have dropped out of college.

Graduate School Education

In addition to reducing undergraduate college enrollment, parents' house values also decrease enrollment in graduate degree programs. The estimated impact of parents' house values on graduate school enrollment is negative and statistically significant at least at a 90% level across the age profile, and is largest at ages 26-27, when a 10% increase in parents' house values during adolescence decreases graduate school enrollment by 0.35 p.p. (4.4%). However, estimates for *any* graduate school attainment are small and statistically indistinguishable from 0 at ages 24-25 and 26-27.

Alternative Specifications and Robustness of Other College Outcomes

Less restrictive specifications and robustness checks reinforce the results that children whose households suffer larger wealth shocks substitute into two-year degree programs from four-year degree programs, and are more likely to drop out of college or to be enrolled in college during their late 20s. Appendix Figure C.1 compares baseline results to less restrictive specifications of Equation (3) and Appendix Figure C.2 compares the baseline results to estimates from samples constructed using alternative cohorts in the control group. The sign and significance of all coefficient estimates are identical when the outcomes are degree non-completion, any college attendance, current undergraduate or graduate enrollment, or any graduate school. When the outcome is associate's degree attainment, the results are even

stronger than those from the baseline specification and sample. The coefficient estimates at ages 22-23 are larger in magnitude (i.e. more negative) and statistically significant at the 95% level of confidence when using alternative samples in the robustness checks. The age 22-23 coefficient estimates are also more negative and statistically significant at a 95% level when estimated using less restrictive specifications of Equation (3). In fact, all estimates across the age profile from the less restrictive specifications are more negative than their baseline counterparts, and the estimate at ages 24-25 is statistically significant at a 95% level when using a specification without any ZIP code controls. Results from alternative specifications and from alternative samples checking for robustness strengthen the evidence that children of parents suffering larger resource shocks during the recession initially substitute into cheaper two-year degree programs.

5.5 Impact of Changes to Parents' House Values on Earnings, Employment, Migration, and Parental Co-Residence

Earnings

Another major finding of this paper is that parents' house values during the Great Recession increased the future earnings of their children. Figure 8 displays the impact of changes to parents' house values during the Great Recession on annual earnings. The timing and persistence of the influence of house values on earnings is identical to the timing and persistence of its influence on bachelor's degree attainment. A 10% increase in parents' house values increases annual earnings by $322 (1.28%) at ages 24-25 and $399 (1.24%) at ages

26-27. Similar to estimates when the outcome is bachelor's degree attainment, less restrictive specifications of Equation (3) produce larger estimates. The coefficient estimate at ages 26-27 when using the least restrictive specification is $483 (1.50%) (See Appendix Figure C.4).

An identification concern that arises when earnings is the outcome is that the outcomes for some members of the control cohort are measured during the Great Recession. This could bias the estimated impacts on earnings if the recession differentially impacts the earnings of individuals in the control group who grew up in ZIP codes that had high or low house value shocks. The expected direction of the resulting bias is downwards: baseline college education levels in high shock ZIP codes are lower than those in low shock ZIP codes (see Appendix Figure C.3), and recessions have smaller effects on annual earnings for those with college degrees (Schwandt and von Wachter, 2019). The implication is that the estimated initial difference between high and low shock ZIP codes is greater than it would be in the absence of the recession, which means the estimated impact of growing up in a low shock ZIP code (and hence greater parents' house values) will likely be too conservative.

Contextualizing Magnitude of Earnings Estimates

The estimated impacts of parents' house values on earnings in this paper are slightly larger and statistically indistinguishable from than those in Stuart (2022) when re-scaling to a comparable treatment dosage. Stuart (2022) finds that a 10% decrease in earnings per capita decreases earnings by 3.3% for children aged 11-19 during the double-dip recession. I find that children from households suffering the average house value decline for ZIP codes

experiencing above median recession house value shocks have earnings reductions of 3.8% in the most restricted specification. The coefficient estimate from the least restricted specification indicates that parents' house value shocks decrease earnings by 4.6%. The estimate from Stuart (2022) is within the 95% confidence interval of the estimates in this paper. It is notable that the estimates in my paper are as large as the estimates in Stuart (2022), for the same reasons that it was surprising that bachelor's attainment estimates were as large (see Section 5.1). Furthermore, it is plausible that the impact of parents' house values on earnings will continue to grow over time due to the fact that parents' resources also increase B.A. attainment, the returns to which grow over the life-cycle (Bhuller et al., 2017).

Employment and Migration

Table 3 reports results for additional labor-market-related outcomes. These results show that parents' house values increase employment, increase migration, and reduce the likelihood that children live with their parents during adulthood. The influence of parents' house values on full-time employment and migration follows the same pattern as its impact on bachelor's degree attainment. The impact at ages 22-23 is statistically indistinguishable from 0 for both outcomes. Estimates at ages 24-25 and 26-27 are positive and statistically significant for both outcomes. A 10% increase in parents' house values increases full-time employment in the prior year by 0.51 p.p. (0.84%) at ages 26-27, and it increases migration to a labor market in a different state by 0.44 p.p. (2.16%) at ages 26-27. These results imply that parents' home values during the Great Recession increased the future employment and migration of their children.

Parental Co-Residence

The final labor-market-related result is that parents' house values during adolescence reduce the likelihood that their children continue to live at home across the entire studied age profile. The estimated impact is largest at ages 26-27, when a 10% increase in parents' house values during adolescence reduces parental co-residence by 1.14 p.p. (4.08%). The fact that parents' resources during adolescence have a substantial effect on parental co-residence is complementary to the full set of results presented in this paper. Children of parents suffering larger home value shocks are less likely to be employed, have lower income, and have less human capital since they are more likely to have left college without a degree. Children use parental co-residence as informal insurance against labor market risk (Kaplan, 2012), and children of parents who suffered larger resource shocks experience worse labor market outcomes. Furthermore, the importance of the parental co-residence insurance channel could be amplified for households who suffered larger shocks to house values during the recession, since these households would have fewer financial resources at their disposal to insure against the income risk of their children (Boar, 2021).

Robustness of Labor Market Outcomes

The estimated impact of parents' house values on earnings, employment, out-of-state migration, and parental co-residence is robust to using samples that include alternative cohorts in the control group. Appendix Figure C.5 compares the estimates when using the main analysis sample to coefficient estimates from Equation (3) that use either the 2000-2002 cohorts or the 2001-2003 cohorts as a control group. The sign and significance of nearly all

coefficient estimates using alternative samples are identical to those from the main analysis sample. The lone exception is the estimated impact of parents' house values on out-of-state migration at ages 22-23 when using the 2000-2002 cohorts as a control group; adding the 2000 cohort to the control group yields a positive estimate of 0.24 p.p. that is significant at a 90% level of confidence.

5.6 Examining the Impact of Parents' House Values on Earnings

The age profile timing at which parents' house values increases earnings is identical to the timing at which it increases bachelor's degree attainment. Parents' house values during adolescence do not impact bachelor's degree attainment or earnings at ages 22-23, when many students are still completing their undergraduate studies, but do increase both bachelor's degree attainment and earnings at ages 24-25 and 26-27. Furthermore, the magnitude of the impact of parents' house values on both outcomes is fairly similar. I perform a mediation exercise to explore the importance of bachelor's degree attainment in explaining the positive impact of parents' house values on earnings. I add the indicator for bachelor's degree attainment as an independent variable to Equation (3) and then estimate the model. The coefficient estimate of parents' house values on earnings should decrease towards 0 when B.A. attainment is included as an endogenous control variable if parents' resources are increasing earnings via an increase in bachelor's degree attainment.

Figure 9 reports the results of this mediation exercise. Including B.A. attainment as a control variable to Equation (3) decreases the estimated impact of parents' house values on future earnings, although the estimates are still substantial and statistically significant. The

coefficient estimate at ages 24-25 drops from $322 in the baseline specification to $258 when B.A. attainment is included as a control, and the estimate at age 26-27 drops from $399 to $354. These results provide suggestive evidence that parents' resources influence earnings via other channels than bachelor's degree attainment alone.

I have also shown in this paper that the value of the parents' housing asset during adolescence persistently increases migration to labor markets out-of-state and full-time employment. Along with bachelor's degree attainment, employment and migration could also drive the increase in earnings. I perform two more mediation exercises to understand how parents' resources increase earnings. First, I estimate Equation (3) with earnings as the dependent variable, adding the indicators for B.A. attainment, migration, and full-time employment as control variables. Next, I estimate Equation (3) with earnings as the outcome, fully interacting bachelor's degree attainment, migration, and full-time employment and adding the products as control variables. The baseline results are compared to the three mediation exercises in Figure 10.

Estimates of parents' house values on earnings at ages 24-25 and 26-27 become smaller when adding endogenous independent variables to Equation (3), although all estimates are still statistically significant at a 95% level of confidence. The mediation exercise that fully interacts B.A. attainment, employment, and migration reduces the estimated impact of parents' house values on earnings at age 26-27 to $180 from its baseline estimate of $399. This mediation exercise provides suggestive evidence that some of the impact of parents' house values on future earnings is due to its effect on educational attainment, employment, and migration. However, the fact that estimates are still positive and statistically significant sug-

gests that parents' house values impact earnings through a channel that I have not explored.

An additional explanation is that parents' house values increase future earnings of college graduates by improving the quality of the college that their children attend. Prior literature demonstrates that parents' resources increase college quality (e.g. Belley and Lochner (2007); Lovenheim and Reynolds (2013)), which in turn increases future earnings (e.g. Black and Smith (2006); Andrews et al. (2014)). If parents' house values during the Great Recession increase college quality, this could increase the future earnings of their children conditional on bachelor's degree attainment. I am unable to mediate earnings with college quality in this paper because the ACS does not include information about the institution attended by college graduates.

6 Conclusion

In this paper, I provide new causal evidence of the intergenerational transmission of recession shocks to children, focusing on the impact of changes to parents' house values during adolescence. Using restricted-access microdata from the U.S. Census Bureau, I create intergenerational linkages between adult children and their parents that include precise information on household location over time. I estimate a generalized difference-in-differences specification that leverages within labor market variation in house values and find that exposure to the average house value decrease during the Great Recession reduced bachelor's degree attainment at ages 26-27 by 0.63 p.p. (1.26%), annual earnings by \$630 (1.95%), and full-time employment by 0.81 p.p. (1.32%). These estimates, while substantial on their

own, understate the effect of parents' resources during the Great Recession. This is because these estimates isolate the impact of parents' house values only. Other shocks to parents' resources during the recession are also likely relevant.

I also find that parents' resources reduce substitution into cheaper two-year degree programs, reduce the likelihood of dropping out of college, reduce enrollment in college programs in a child's late 20s, increase migration to different labor markets, and reduce the take-up of informal insurance through parental co-residence. Finally, I also provide suggestive evidence that estimates of the impact of parents' house values from an empirical approach that does not leverage within labor market variation in house values are biased.

There are a pair of important takeaways from the findings in my paper. The first is that the value of the parents' housing asset has a substantial impact on the future college education attainment and earnings outcomes of their children. The fact that I study parents' house value shocks during adolescence means that the results are likely driven by parents financing college education. Additionally, these results demonstrate that the negative long-run consequences of recessions include impacts on children who are not yet in the labor market. Specifically, recessions reduce the college attainment and future earnings of those who are adolescents during the shocks.

The results of this paper have significant policy relevance. The fact that parents' house values impact the outcomes of their children in adulthood should be taken into account when designing housing policy. For example, there is causal evidence that states without a judicial requirement for foreclosures had higher disclosure rates and substantially steeper home price declines during the Great Recession (Mian et al., 2015). The results presented in

this paper imply that a spillover of this housing policy choice is that judicial requirements for foreclosure impact college attainment via its impact on the value of parents' housing assets. Furthermore, my results suggest that it could be worthwhile to include adjustments to higher education policies when implementing countercyclical policy measures. Policies could include increases to federal financial aid through grants or borrowing limits, or continued emphasis on income-driven repayment policies that reduce income risk associated with investments in higher education. This paper does not determine the precise mechanism through which decreases to parents' house values decreased bachelor's degree attainment and earnings, and therefore it is beyond the scope of this paper to suggest precise policy prescriptions. Promising areas for future research include determining the mechanism through which house values impact the college outcomes of children, and whether these results generalize to later cohorts. Later cohorts are particularly interesting, given that they have been treated by recession shocks during childhood and house price appreciation during adolescence.

A Appendix

A.1 Matching Procedure of Analysis Sample to MAF-ARF

The purpose of matching my sample to the MAF-ARF is to observe where households lived during 2007, at the beginning of the Great Recession. With this in mind, I use the following procedure to link children and their parents to the MAF-ARF. I begin by linking individuals to the 2007 MAF-ARF. If I am able to link individuals to a MAFID that matches their MAFID in the 2000 Decennial Census, that means that the household lives at the same

address, and I use this MAFID as their location during the recession. If I am not able to match individuals to their original MAFID, but instead match them to a unique MAFID alongside their parents in the 2007 MAF-ARF, I use this MAFID to assign their location during the recession. If I am unable to assign an individual to a location using these first two criteria, I proceed by taking advantage of the fact that I have an identifier for both parents and children. Specifically, if a child and/or parent is matched to multiple MAFIDs, and I am able to match a child/parent pair to the same MAFID, I use this mutually matching MAFID to assign a geographic location during the recession. If an individual is still unmatched to a MAFID, I am unable to use the 2007 MAF-ARF to assign them to a geographic location during the Great Recession. I take these unmatched individuals, and attempt to match them to a MAFID in the 2008 MAF-ARF using the same 3-step procedure described above. If that fails, I then proceed to the 2009 and then the 2010 MAF-ARF.

A.2 Construction of Selected Control Variables

I use the ACS and the Decennial Census to construct the household and individual controls that I include in my specifications and will describe in this section. White is an indicator that is equal to 1 if race is white alone and ethnicity is non-Hispanic. Hispanic if is an indicator that is equal to 1 if ethnicity is Hispanic. Black is an indicator that is equal to 1 if race is Black alone and ethnicity is non-Hispanic. Asian is an indicator that is equal to 1 if race is Asian alone and ethnicity is non-Hispanic. All other race/ethnicity combinations are considered "other" and serve as an omitted variable in regression specifications. Information used to construct the race/ethnicity indicators comes from the ACS.

Female is an indicator equal to 1 if sex is female in the ACS. Parent Married 2000 is an indicator equal to 1 if the parent householder in the 2000 Decennial Census has a spouse. Householder 2000 Male is an indicator equal to 1 if the parent householder in the 2000 Decennial Census is a male. Biological Child is an indicator equal to 1 if the child's relationship to the householder is a biological child in the 2000 Decennial Census. Adopted Child is an indicator equal to 1 if the child's relationship to the householder is an adopted child in the 2000 Decennial Census. The omitted child-relationship category is step-child child.

A.3 Household Internalization of Local Housing Value Changes

Households internalized local house value changes as real changes to value of their housing asset during the Great Recession. Using the 2009-2013 waves of the restricted-access Panel Study of Income Dynamics (PSID) matched to ZIP code home values from Zillow, I estimate the following simple regression:

$$\Delta HV_{h,t} = \pi Pred\Delta HV_{h,t,z(h)} + \epsilon_{h,t}. \tag{6}$$

The dependent variable, $\Delta HV_{h,t}$, is the change in household h's self-assessed value of their home between year t and the previous survey wave[16]. The independent variable, $Pred\Delta HV_{h,t,z(h)}$, is the predicted change in household given the change in ZIP code z's home value index between survey waves. Specifically, $Pred\Delta HV_{h,t,z(h)} = ((\frac{ZHVI_z^t - ZHVI_z^{t-1}}{ZHVI_z^{t-1}}) + 1) * HV_{h,(t-1)}$, where

[16]The PSID is a bi-annual survey

$ZHVI_z^t, ZHVI_z^{t-1}$ are the value of the Zillow Home Value Index in household h's ZIP code z at survey wave t and in the previous survey wave $t-1$, respectively, and $HV_{h,(t-1)}$ is h's self-assessed house value from the previous survey wave. The coefficient π represents the relationship between changes in local house values and self-assessed house values. In particular, an estimate of 1 would imply that a $1 increase in local house values is associate with a $1 increase in a homeowner's self-assessed house value. I cluster standard errors by household, as there is likely serial correlation in house values. Appendix Table B.9 reports the results from Estimating Equation B.9.

The coefficient estimate of 0.57 means that $1 increase in predicted house values results in a $0.57 increase in self-assessed house values. This precise estimate that households internalize local house value changes as changes to the value of their housing asset, but at less than a 1-1 ratio, is consistent with Davis and Quintin (2017). These authors also find that households internalize house value changes but at less than a 1-to-1 ratio.

A.4 Robustness Using the 2001-2004 Age-18 Cohorts as the Control Group

In addition to the robustness checks presented in Section 5.3, I also perform a robustness exercise that uses the 2001-2004 birth year cohorts as the control group. The result that parents' house values lead to a persistent increase in bachelor's degree attainment is sensitive to the inclusion of the 2003 and 2004 cohorts as the control group when using the most restrictive specification in terms of parametric ZIP controls. Results when using less

restrictive specifications of Equation 3 continue to be robust. Appendix Figure C.6 reports the estimates from specifications of Equation (3) that differ by the parametric ZIP controls included as independent variables, estimated using the 2001-2004 age-18 year cohorts as the control group. The pattern from the coefficient estimates indicate that the value of parents' housing asset delays graduation, but leads to persistent increases at later ages. Coefficient estimates are also reported in Appendix Table B.10. Coefficient estimates for the other outcomes are reported in Appendix Table B.11. The other outcomes are also robust to using the 2001-2004 cohorts as a control group.

The issue with including the 2004 age-18 cohort in the control group is that they are likely treated by the Great Recession. This is due to the fact that many members of this cohort are likely to be still in college when their parents' experience resource shocks from the Great Recession. The results presented in this paper consistently show that parents' house values does not begin to increase bachelor's degree attainment until the ages of 24-25. Members of the 2004 age-18 cohort do not turn 24 until 2010, i.e. after the end of the Great Recession. If children of the 2004 cohort from households experiencing relatively larger wealth shocks are less likely to complete college as a result their parents experiencing a shock to the value of their home, than the average estimated "initial difference" between low shock and high shock households would be larger than it would be in a scenario in which the control group did not experience treatment. The expected direction of the resulting bias from including later cohorts in the control group is negative. The evidence presented by this robustness check suggests that this is the case; including plausibly treated cohorts in the control groups shrinks the size of the estimated impact of parents' house values on B.A. attainment relative

to the baseline estimates.

References

Aladangady, A. (2017). Housing Wealth and Consumption: Evidence from Geographically Linked Microdata. *The American Economic Review*, 107(11):3415–3446.

Altonji, J. G., Kahn, L. B., and Speer, J. D. (2016). Labor Market Conditions, Field of Study, and Career Success. *Journal of Labor Economics*, 34(1):362–401.

Andrews, R. J., Li, J., and Lovenheim, M. F. (2014). Quantile Treatment Effects of College Quality on Earnings. *Journal of Human Resources*, 51(1):201–237.

Anstreicher, G. and Miller, L. (2023). Who Scars the Easiest? College Quality and the Effects of Graduating into a Recession. Working paper, Annenberg Institute at Brown University.

Barr, A. and Turner, S. E. (2013). Expanding Enrollments and Contracting State Budgets: The Effect of the Great Recession on Higher Education. *The Annals of the American Academy of Political and Social Science*, 650:168–193.

Bastian, J. and Michelmore, K. (2018). The Long-Term Impact of the Earned Income tax Credit on Children's Education and Employment Outcomes. *Journal of Labor Economics*, 36(4):1127–1163.

Becker, G. S. (1962). Investment in Human Capital: A Theoretical Analysis. *Journal of Political Economy*, 70(5):9–49.

Belley, P. and Lochner, L. (2007). The Changing Role of Family Income and Ability in Determining Educational Achievement. *Journal of Human Capital*, 1(1):37–89.

Bettinger, E., Gurantz, O., Kawano, L., Sacerdote, B., and Stevens, M. (2019). The Long-Run Impacts of Financial Aid: Evidence from California's Cal Grant. *American Economic Journal: Economic Policy*, 11(1):64–94.

Betts, J. R. and McFarland, L. L. (1995). Safe Port in a Storm: The Impact of Labor Market Conditions on Community College Enrollments. *Journal of Human Resources*, 30(4):741–765.

Bhuller, M., Mogstad, M., and Salvanes, K. G. (2017). Life-Cycle Earnings, Education Premiums, and Internal Rates of Return. *Journal of Labor Economics*, 35(4):993–1030.

Black, D. A. and Smith, J. A. (2006). Estimating the Returns to College Quality with Multiple Proxies for Quality. *Journal of Labor Economics*, 24(3):701–728.

Black, S. E., Denning, J. T., Dettling, L. J., Goodman, S., and Turner, L. J. (2020). Taking it to the Limit: Effects of Increased Student Loan Availability on Attainment, Earnings, and Financial Well-Being. Working Paper 27658, National Bureau of Economic Research.

Blom, E., Cadena, B. C., and Keys, B. J. (2021). Investments over the Business Cycle: Insights from College Major Choice. *Journal of Labor Economics*, 39(4):1043–1082.

Boar, C. (2021). Dynastic Precautionary Savings. *Review of Economic Studies*, 88(6):2735–2765.

Bulman, G., Fairlie, R., Goodman, S., and Isen, A. (2021). Parental Resources and College Attendance: Evidence from Lottery Wins. *American Economic Review*, 111(4):1201–1240.

Charles, K. K., Hurst, E., and Notowidigdo, M. J. (2018). Housing Booms and Busts, Labor Market Opportunities, and College Attendance. *American Economic Review*, 108(10):2947–2994.

Chetty, R., Friedman, J. N., and Rockoff, J. (2011). New Evidence on the Long-Term Impacts of Tax Credits. Technical report, Internal Review Service.

Chetty, R., Friedman, J. N., and Rockoff, J. E. (2014a). Measuring the Impacts of Teachers II: Teacher Value-Added and Student Outcomes in Adulthood. *American Economic Review*, 104(9):2633–2679.

Chetty, R., Hendren, N., and Katz, L. F. (2016). The Effects of Exposure to Better Neighborhoods on Children: New Evidence from the Moving to Opportunity Experiment. *American Economic Review*, 106(4):855–902.

Chetty, R., Hendren, N., Kline, P., and Saez, E. (2014b). Where is the Land of Opportunity? The Geography of Intergenerational Mobility in the United States. *The Quarterly Journal of Economics*, 129(4):1553–1623.

Chetty, R., Hendren, N., Kline, P., Saez, E., and Turner, N. (2014c). Is the United States Still a Land of Opportunity? Recent Trends in Intergenerational Mobility. *American Economic Review: Papers & Proceedings*, 104(5):141–147.

Christian, M. S. (2007). Liquidity Constraints and the Cyclicality of College Enrollment in the United States. *Oxford Economic Papers*, 59(1):141–169.

Chyn, E. (2018). Moved to opportunity: The long-run effects of public housing demolition on children. *American Economic Review*, 108(10):3028–3056.

Cooper, D. and Luengo-Prado, M. J. (2015). House Price Growth when Children are Teenagers: A Path to Higher Earnings? *Journal of Urban Economics*, 86:54–72.

Davis, M. and Quintin, E. (2017). On the Nature of Self-Assessed House Prices. *Real Estate Economics*, 45(3):628–649.

Dellas, H. and Sakellaris, P. (2003). On the Cyclicality of Schooling: Theory and Evidence. *Oxford Economics Papers*, 55:148–172.

Dynarski, S. M. (2008). Building the Stock of College-Educated Labor. *Journal of Human Resources*, 43(3):576–610.

Dynarski, S. M. and Bailey, M. (2011). Gains and Gaps: Changing Inequality in US College Entry and Completion. In Duncan, G. and Murnane, R., editors, *Whither Opportunity: Rising Inequality, Schools, and Children's Life Chances*. Russell Sage.

Gilraine, M., Graham, J., and Zhang, A. (2023). Public Education and Intergenerational Housing Wealth Effects. Working Paper 31345, National Bureau of Economic Research.

Hilger, N. G. (2016). Parental Job Loss and Children's Long-Term Outcomes: Evidence from 7 Million Fathers' Layoffs. *American Economic Journal: Applied Economics*, 8(3):247–283.

Hotz, V. J., Wiemers, E. E., Rasmussen, J., and Koegel, K. M. (2023). The Role of Parental Wealth and Income in Financing Children's College Attendance and Its Consequences. *Journal of Human Resources*, 58(6).

Kahn, L. B. (2010). The Long-Term Labor Market Consequences of Graduating from College in a Bad Economy. *Labour Economics*, 17:303–316.

Kaplan, G. (2012). Moving Back Home: Insurance against Labor Market Risk. *Journal of Political Economy*, 120(2):446–512.

Lochner, L. and Monge-Naranjo, A. (2011). The Nature of Credit Constraints and Human Capital. *American Economic Review*, 101(6):2487–2529.

Lochner, L. and Monge-Naranjo, A. (2012). Credit constraints in education. *Annual Review of Economics*, 4(1):225–256.

Long, B. T. (2015). The Financial Crisis and Declining College Affordability: How Have Students and their Families Respondent? In Brown, J. R. and Hoxby, C. M., editors, *How the Great Recession Affected Higher Education*. University of Chicago Press, Chicago, IL.

Lovenheim, M. F. (2011). The Effect of Liquid Housing Wealth on College Enrollment. *Journal of Labor Economics*, 29(4):741–771.

Lovenheim, M. F. and Reynolds, C. L. (2013). The Effect of Housing Wealth on College Choice: Evidence from the Housing Boom. *Journal of Human Resources*, 48(1):1–35.

Manoli, D. and Turner, N. (2018). Cash-on-Hand and College Enrollment: Evidence from

Population Tax Data and the Earned Income Tax Credit. *American Economic Journal: Economic Policy*, 10(2):242–271.

Mian, A., Rao, K., and Sufi, A. (2013). Household Balance Sheets, Consumption, and the Economic Slump. *Quarterly Journal of Economics*, 128:1687–1726.

Mian, A. and Sufi, A. (2009). The Consequences of Mortgage Credit Expansion: Evidence from the U.S. Mortgage Default Crisis. *Quarterly Journal of Economics*, 124:1449–1496.

Mian, A. and Sufi, A. (2011). House Prices, Home Equity-Based Borrowing, and the US Hosuehold Leverage Crisis. *American Economic Review*, 101:2132–2156.

Mian, A. and Sufi, A. (2014). *House of Debt*. The University of Chicago Press.

Mian, A., Sufi, A., and Trebbi, F. (2015). oreclosures, House Prices, and the Real Economy. *The Journal of Finance*, 70:2587–2634.

Mumford, K. J. (2022). Student Selection into an Income Share Agreement. Working paper, Annenberg Institute at Brown University.

Oreopoulos, P., von Wachter, T., and Heisz, A. (2012). The Short- and Long-Term Career Effects of Graduating in a Recession. *American Economic Journal: Applied Economics*, 4(1):1–29.

Rinz, K. (2022). Did Timing Matter? Life Cycle Differences in Effects of Exposure to the Great Recession. *Journal of Labor Economics*, 40(3):703–735.

Rothstein, J. (2020). The lost generation? labor market outcomes for post great recession entrants. Working Paper 27516, National Bureau of Economic Research.

Sakellaris, P. and Spilimbergo, A. (2000). Business Cycles and Investment in Human Capital: International Evidence on Higher Education. *Carnegie-Rochester Conference Series on Public Policy*, 52:221–256.

Schwandt, H. and von Wachter, T. (2019). Unlucky Cohorts: Estimating the Long-Term Effects of Entering the Labor Market in a Recession in Large Cross-Sectional Datasets. *Journal of Labor Economics*, 37(1):161–198.

Steven Manson, Jonathan Schroeder, David Van Riper, Tracy Kugler, and Steven Ruggles (2022). IPUMS National Historical Geographic Information System: Version 17.0 [dataset].

Stuart, B. A. (2022). The Long-Run Effects of Recessions on Education and Income. *American Economic Journal: Applied Economics*, 14(1):42–74.

Sullivan, J. and Genadek, K. (2022). Using the Census Bureau's Master Address File for Migration Research.

Yagan, D. (2019). Employment Hysteresis from the Great Recession. *Journal of Political Economy*, 127(5):2505–2558.

B Tables

Table 1: Summary Statistics Describing the Outcomes of the Main Analysis Sample

	Age Bin		
	22-23	24-25	26-27
B.A. Attainment	0.314	0.457	0.499
A.A. Attainment	0.091	0.091	0.090
Degree Non-completion	0.144	0.158	0.159
Any College	0.750	0.784	0.796
Current U.G. Enrollment	0.358	0.147	0.093
Earnings	$14760	$25070	$32260
Full-Time Employment	0.314	0.527	0.610
Current Grad Enrollment	0.066	0.094	0.081
Any Graduate School	0.072	0.133	0.170
Parental Co-residence	0.517	0.385	0.279
Out-of-State Migration	0.128	0.178	0.205

Notes: This table reports the averages of the eleven outcome variables studied in this paper. Averages are reported for each of the three age bins that I study. The values in the first column report the average of the given outcome variables for members of the main analysis sample who are between the ages of 22-23. The second column reports average outcomes for individuals aged 24-25, and the third column reports outcomes for individuals aged 26-27. The age bins pool the treatment and control groups together. The total sample size is 937,000.

Table 2: Impact of Parents' House Values on Other College Education Outcomes

	Dependent Variable:					
	Any College	A.A. Attainment	Degree Non-completion	Current U.G. Enrollment	Current Grad Enrollment	Any Graduate School
Age 22-23	-0.0076***	-0.0019*	-0.0037***	0.0010	-0.0017*	-0.0036***
	(0.0017)	(0.0011)	(0.0014)	(0.0015)	(0.0010)	(0.0011)
Age 24-25	-0.0059***	-0.0012	-0.0043***	-0.0079***	-0.0018*	-0.0011
	(0.0017)	(0.0011)	(0.0014)	(0.0015)	(0.0010)	(0.0011)
Age 26-27	-0.0053***	-0.0000	-0.0059***	-0.0049***	-0.0035***	-0.0005
	(0.0017)	(0.0011)	(0.0014)	(0.0015)	(0.0010)	(0.0012)
N	937000	937000	937000	937000	937000	937000
Adj. R^2	0.10	0.02	0.02	0.12	0.03	0.06

Notes: Each column displays the coefficient estimates of β_j from Equation (3). See Section 3.2 for definition of outcome variables. Standard errors are clustered by ZIP code. * if $p < 0.1$, ** if $p < 0.05$, *** if $p < 0.01$.

Table 3: Impact of Parents' House Values on Employment, Parental Co-residence, and Migration

	Dependent Variable:		
	Employment	Parental Co-residence	Migration Out-of-State
Age 22-23	-0.0011	-0.0061***	0.0010
	(0.0018)	(0.0019)	(0.0014)
Age 24-25	0.0035*	-0.0092***	0.0034**
	(0.0018)	(0.0019)	(0.0014)
Age 26-27	0.0051***	-0.0114***	0.0044***
	(0.0019)	(0.0019)	(0.0014)
N	937000	937000	908000
Adj. R^2	0.11	0.09	0.12

Notes: Each column displays coefficient estimates of β_j from Equation (3). See Section 3.2 for definition of outcome variables. The ACS does not contain standard geographic information for individuals who live in group quarters, which is why the sample size for the migration outcome is slightly smaller than the other sample sizes. Standard errors are clustered by ZIP code. * if $p < 0.1$, ** if $p < 0.05$, *** if $p < 0.01$.

Table B.1: Summary Statistics Describing the Main Analysis Sample

	By Cohort Type		
	Control	Treat	Full Sample
Outcomes:			
B.A. Attainment	0.379	0.434	
A.A. Attainment	0.091	0.091	
Degree Non-completion	0.160	0.151	
Any College	0.751	0.785	
Current U.G. Enrollment	0.212	0.201	
Earnings	$23740	$23710	
Full-Time Employment	0.468	0.482	
Current Grad Enrollment	0.081	0.080	
Any Graduate School	0.120	0.125	
Parental Co-residence	0.362	0.411	
Out-of-State Migration	0.156	0.167	
Selected Controls:			
Parent Married 2000	0.837	0.892	0.878
Move	0.105	0.233	0.200
Householder 2000 Male	0.789	0.815	0.808
Biological Child	0.907	0.942	0.933
Adopted Child	0.026	0.025	0.025
White	0.772	0.760	0.763
Hispanic	0.101	0.113	0.110
Asian	0.040	0.041	0.040
Black	0.064	0.0542	0.057
Female	0.500	0.493	0.495
N (By Treat and Control Only)	243000	694000	937000

Notes: This table displays the mean of the outcomes and selected control variables from the main analysis sample. The column titled "Control" reports the mean of outcomes and controls for sample respondents who are part of the control group. These respondents who turn 18 in the years 2001-2002. The column "Treat" reports the mean for outcomes and controls for respondents who turn 18 in the years 2007-2012. The full sample reports the mean of control variables for the entire sample.

Table B.2: Coefficient Estimates of the Impact of Parents' House Values on B.A. Attainment: ZIP Code Specifications

	(1)	(2)	(3)	(4)	(5)
Age 22-23	-0.00183	-0.00074	0.00032	-0.00296*	-0.00218
	(0.00171)	(0.00168)	(0.00160)	(0.00154)	(0.00154)
Age 24-25	0.00571***	0.00679***	0.00785***	0.00380**	0.00504***
	(0.00171)	(0.00168)	(0.00161)	(0.00154)	(0.00154)
Age 26-27	0.00399**	0.00507***	0.00613***	0.00272*	0.00340**
	(0.00175)	(0.00172)	(0.00164)	(0.00157)	(0.00158)
N	937000	937000	937000	1047000	1066000
Adj. R^2	0.19	0.19	0.19	0.19	0.19
Sample	Main Sample	Main Sample	Main Sample	2000-2002 Cohorts	2001-2003 Cohorts
ZIP Controls	Full Controls	Dem. Controls Only	No Controls	Full Controls	Full Controls

Notes: Each column displays the coefficient estimates of β_j from a specification of Equation (3) when bachelor's degree attainment is the outcome. Column (1) reports estimates from the main specification, represented by Figure 3. Column (2) reports estimates from the specification 3 that only includes ZIP controls for changes in education attainment and race shares. Column (3) reports estimates from the specification of Equation (3) that does not include any ZIP controls. Columns (1)-(3) are also reported in Figure 4, and all are estimated using the baseline sample. Column (4) reports estimates from Equation (3) estimated using a sample that adds the 2000 year-18 cohort to the control group. Column (5) reports estimates from Equation (3) estimated using a sample that adds the 2003 year-18 cohort to the control group. Columns (4)-(5) are the robustness checks, and are reported with column (1) in Figure 7. Standard errors are clustered by ZIP code. * if $p < 0.1$, ** if $p < 0.05$, *** if $p < 0.01$.

Table B.3: Coefficient Estimates of the Impact of Parents' House Values on B.A. Attainment: Metro Specifications

	(1)	(2)	(3)
Age 22-23	-0.00135	-0.00184	-0.00278*
	(0.00168)	(0.00162)	(0.00156)
Age 24-25	0.00408**	0.00276*	0.00237
	(0.00159)	(0.00156)	(0.00145)
Age 26-27	0.00187	0.00127	0.00023
	(0.00163)	(0.00163)	(0.00150)
N	937000	1047000	1066000
Adj. R^2	0.18	0.18	0.18
Sample	Main Sample	2000-2002 Cohorts	2001-2003 Cohorts

Notes: Each column displays the coefficient estimates of λ_j from Equation (4) when bachelor's degree attainment is the outcome. Column (1) reports results that were estimated using the main analysis sample. Results in column (2) are estimated using a sample that adds the year-2000 age-18 cohort to the control group. Results in column (3) are estimated using a sample that adds the year-2003 age-18 cohort to the control group. Column (1) is reported in Figure 5, Columns (1)-(3) are reported in Figure 7. Standard errors are clustered by metro area. * if $p < 0.1$, ** if $p < 0.05$, *** if $p < 0.01$.

Table B.4: Coefficient Estimates of Parents' House Values on Other College Education Outcomes by ZIP Controls

Panel A: Demographic ZIP Controls Only

	Dependent Variable:					
	Any College	A.A. Attainment	Degree Non-completion	Current U.G. Enrollment	Current Grad Enrollment	Any Graduate School
Age 22-23	-0.00762***	-0.00250**	-0.00431***	0.00106	-0.00189**	-0.00385***
	(0.00167)	(0.00105)	(0.00139)	(0.00152)	(0.00099)	(0.00114)
Age 24-25	-0.00588***	-0.00168	-0.00486***	-0.00778***	-0.00193**	-0.00131
	(0.00165)	(0.00105)	(0.00138)	(0.00145)	(0.00100)	(0.00114)
Age 26-27	-0.00535***	-0.00060	-0.00649***	-0.00479***	-0.00372***	-0.00071
	(0.00166)	(0.00106)	(0.00139)	(0.00146)	(0.00101)	(0.00118)
N	937000	937000	937000	937000	937000	937000
Adj. R^2	0.10	0.02	0.02	0.11	0.03	0.06

Panel B: No ZIP Controls

	Dependent Variable:					
	Any College	A.A. Attainment	Degree Non-completion	Current U.G. Enrollment	Current Grad Enrollment	Any Graduate School
Age 22-23	-0.00831***	-0.00331***	-0.00488***	0.00049	-0.00255***	-0.00457***
	(0.00160)	(0.00100)	(0.00133)	(0.00145)	(0.00095)	(0.00109)
Age 24-25	-0.00658***	-0.00248**	-0.00543***	-0.00835***	-0.00260***	-0.00204*
	(0.00157)	(0.00100)	(0.00131)	(0.00138)	(0.00095)	(0.00110)
Age 26-27	-0.00604***	-0.00141	-0.00706***	-0.00537***	-0.00438***	-0.00143
	(0.00158)	(0.00101)	(0.00132)	(0.00138)	(0.00097)	(0.00114)
N	937000	937000	937000	937000	937000	937000
Adj. R^2	0.10	0.02	0.02	0.11	0.03	0.06

Notes: Each column displays the coefficient estimates of β_j from specifications of Equation (3) that vary by the ZIP code controls included. See Section 3.2 for definition of outcome variables. Panel A reports estimates from an alternate specification of Equation (3) that includes only parametric ZIP controls for changes in race shares and changes in education shares. Panel B reports estimates from an alternate specification of Equation (3) that does not include any parametric ZIP controls. Standard errors are clustered by ZIP code. Results in Panel A and B are also reported in Appendix Figure C.1. * if $p < 0.1$, ** if $p < 0.05$, *** if $p < 0.01$.

Table B.5: Coefficient Estimates of Parents' House Values on Other College Education Outcomes, Robustness Checks

Panel A: 2000-2002 Year-18 Cohorts as Control Group

	Dependent Variable:					
	Any College	A.A. Attainment	Degree Non-completion	Current U.G. Enrollment	Current Grad Enrollment	Any Graduate School
Age 22-23	-0.00827***	-0.00223**	-0.00352***	0.00180	-0.00140	-0.00355***
	(0.00155)	(0.00097)	(0.00128)	(0.00136)	(0.00089)	(0.00102)
Age 24-25	-0.00673***	-0.00133	-0.00413***	-0.00615***	-0.00150*	-0.00121
	(0.00153)	(0.00096)	(0.00126)	(0.00128)	(0.00090)	(0.00103)
Age 26-27	-0.00614***	-0.00028	-0.00587***	-0.00388***	-0.00316***	-0.00037
	(0.00153)	(0.00098)	(0.00128)	(0.00129)	(0.00091)	(0.00106)
N	1047000	1047000	1047000	1047000	1047000	1047000
Adj. R^2	0.10	0.02	0.02	0.11	0.03	0.06

Panel A: 2000-2002 Year-18 Cohorts as Control Group

	Dependent Variable:					
	Any College	A.A. Attainment	Degree Non-completion	Current U.G. Enrollment	Current Grad Enrollment	Any Graduate School
Age 22-23	-0.00734***	-0.00208**	-0.00384***	0.00176	-0.00159*	-0.00386***
	(0.00147)	(0.00097)	(0.00122)	(0.00137)	(0.00088)	(0.00103)
Age 24-25	-0.00560***	-0.00126	-0.00438***	-0.00661***	-0.00154*	-0.00129
	(0.00144)	(0.00095)	(0.00121)	(0.00129)	(0.00088)	(0.00103)
Age 26-27	-0.00521***	-0.00020	-0.00596***	-0.00399***	-0.00340***	-0.00082
	(0.00145)	(0.00097)	(0.00124)	(0.00130)	(0.00090)	(0.00107)
N	1066000	1066000	1066000	1066000	1066000	1066000
Adj. R^2	0.11	0.02	0.02	0.11	0.03	0.06

Notes: Each column displays the coefficient estimates of β_j from Equation (3), estimated using alternative samples. See Section 3.2 for definitions of the outcome variables. Panel A reports estimates from an alternate sample that adds the 2000 age-18 cohort to the control group. Panel B reports estimates from an alternate sample that adds the 2003 age-18 cohort to the control group. Standard errors are clustered by ZIP code. Results in Panel A and B are also reported in Appendix Figure C.2. * if $p < 0.1$, ** if $p < 0.05$, *** if $p < 0.01$.

Table B.6: Coefficient Estimates of the Impact of Parents' House Values on Annual Earnings

	(1)	(2)	(3)	(4)	(5)
Age 22-23	43	38	127	37	11
	(92)	(90)	(86)	(80)	(79)
Age 24-25	322***	316***	406***	305***	287***
	(92)	(90)	(87)	(80)	(79)
Age 26-27	399***	394***	483***	390***	360***
	(96)	(95)	(91)	(85)	(84)
N	937000	937000	937000	1047000	1066000
Adj. R^2	0.11	0.11	0.11	0.11	0.11
Sample	Main Sample	Main Sample	Main Sample	2000-2002 Cohorts	2001-2003 Cohorts
ZIP Controls	Full Controls	Dem. Controls Only	No Controls	Full Controls	Full Controls

Notes: Each column displays the coefficient estimates of β_j from a specification of Equation (3) when annual earnings is the outcome. Column (1) reports estimates from the main specification, represented by Figure 8. Column (2) reports estimates from the specification 3 that only includes ZIP controls for changes in education attainment and race shares. Column (3) reports estimates from the specification of Equation (3) that does not include any ZIP controls. Columns (1)-(3) are also reported in Appendix Figure C.4, and all are estimated using the baseline sample. Column (4) reports estimates from Equation (3) estimated using a sample that adds the 2000 year-18 cohort to the control group. Column (5) reports estimates from Equation (3) estimated using a sample that adds the 2003 year-18 cohort to the control group. Columns (4)-(5) are the robustness checks, and are reported with column (1) in Appendix Figure C.5. Standard errors are clustered by ZIP code. * if $p < 0.1$, ** if $p < 0.05$, *** if $p < 0.01$.

Table B.7: Coefficient Estimates of Parents' House Values on Other Labor Market Outcomes, Robustness Checks

Panel A: 2000-2002 Year-18 Cohorts as Control Group

	Dependent Variable:		
	Employment	Parental Co-residence	Migration Out-of-State
Age 22-23	-0.00083	-0.00578***	0.00241*
	(0.00159)	(0.00170)	(0.00124)
Age 24-25	0.00293*	-0.00953***	0.00479***
	(0.00159)	(0.00168)	(0.00124)
Age 26-27	0.00513***	-0.01110***	0.00575***
	(0.00163)	(0.00173)	(0.00128)
N	1047000	1047000	1016000
Adj. R^2	0.09	0.12	0.06

Panel A: 2000-2002 Year-18 Cohorts as Control Group

	Dependent Variable:		
	Employment	Parental Co-residence	Migration Out-of-State
Age 22-23	-0.00109	-0.00774***	0.00142
	(0.00161)	(0.00161)	(0.00121)
Age 24-25	0.00309*	-0.01117***	0.00372***
	(0.00162)	(0.00161)	(0.00121)
Age 26-27	0.00512***	-0.01298***	0.00476***
	(0.00167)	(0.00166)	(0.00124)
N	1066000	1066000	1032000
Adj. R^2	0.08	0.12	0.06

Notes: Each column displays the coefficient estimates of β_j from Equation (3), estimated using alternative samples. Each column displays coefficient estimates of β_j from Equation (3). See Section 3.2 for definitions of the outcome variables. The ACS does not contain standard geographic information for individuals who live in group quarters, which is why the sample size for the migration outcome is slightly smaller than the other sample sizes. Panel A reports estimates from an alternate sample that adds the 2000 age-18 cohort to the control group. Panel B reports estimates from an alternate sample that adds the 2003 age-18 cohort to the control group. Standard errors are clustered by ZIP code. Results in Panel A and B are also reported in Appendix Figure C.5. * if $p < 0.1$, ** if $p < 0.05$, *** if $p < 0.01$.

Table B.8: Coefficient Estimates of the Regressions Mediating Impact of Parents' House Values on Earnings

	(1)	(2)	(3)
Age 22-23	63	98	138*
	(91)	(79)	(78)
Age 24-25	258***	194**	193**
	(91)	(79)	(78)
Age 26-27	355***	238***	180**
	(95)	(82)	(82)
N	937000	908000	908000
Adj. R^2	0.14	0.32	0.33
Mediators	B.A. Attainment	B.A. Attainment, Employment, Migration	B.A. Attainment, Employment, Migration
Mediators Fully Interacted	N	N	Y

Notes: This table reports the coefficient estimates of parents' house values on earnings from the mediation exercises described in Section 5.6. Column (1) reports the estimates from the mediation exercise that only includes B.A. Attainment as an endogenous control variable. Column (2) reports the estimates from the mediation exercise that includes B.A. attainment, full-time employment, and out-of-state migration as endogenous control variables. Column (3) reports estimates from the mediation exercise that fully interacts B.A. attainment, full-time employment, and out-of-state migration and includes the products as endogenous control variables. The coefficient estimates and standard errors are graphically reported in Figure 10. Standard errors are clustered by ZIP code. * if $p < 0.1$, ** if $p < 0.05$, *** if $p < 0.01$.

Table B.9: Household Internalization of Local Housing Value Changes

	Coefficient Estimate	Standard Error	R^2	N	P-Value
Predicted House Value Change	0.57	(0.02)	0.13	8,841	0.00

Notes: The table reports the output from estimating Equation 6. Data is from the 2009-2013 waves of the PSID matched to the home value data from the Zillow Home Value Index. The column Coefficient Estimate reports the estimate of π. Standard errors are clustered by household. N reports the number of observations used to estimate Equation 6.

Table B.10: Coefficient Estimates of the Impact of Parents' House Values on B.A. Attainment: 2001-2004 Cohorts as Control Group

	(1)	(2)	(3)	(4)
Age 22-23	-0.00347**	-0.00262*	-0.00205	-0.00346*
	(0.00143)	(0.00140)	(0.00133)	(0.00151)
Age 24-25	0.00353**	0.00439***	0.00496***	0.00155
	(0.00143)	(0.00140)	(0.00133)	(0.00139)
Age 26-27	0.00185	0.00290*	0.00327**	-0.00069
	(0.00147)	(0.00145)	(0.00138)	(0.00150)
N	1194000	1194000	1194000	1194000
R^2	0.19	0.19	0.19	0.18
Specification Type	ZIP Code	ZIP Code	ZIP Code	Metro Area
ZIP Controls	Full Controls	Dem. Controls Only	No Controls	N/A

Notes: Each column displays the coefficient estimates of β_j from a specification of Equation (3) when bachelor's degree attainment is the outcome. All results are estimated using an alternative sample that uses the 2001-2004 age-18 cohorts as the control group. Column (1) reports estimates from the main specification. Column (2) reports estimates from the specification 3 that only includes ZIP controls for changes in education attainment and race shares. Column (3) reports estimates from the specification of Equation (3) that does not include any ZIP controls. Columns (1)-(3) are also reported in Figure C.6. Standard errors for the estimates presented in columns (1)-(3) are clustered by ZIP Code. Column (4) reports the coefficient estimates λ_j from Equation (4). These results are also estimated using the alternative sample that uses the 2001-2004 age-18 cohorts are the control group. Standard errors are clustered by metropolitan area. * if $p < 0.1$, ** if $p < 0.05$, *** if $p < 0.01$.

Table B.11: Coefficient Estimates of Parents' House Values on College and Labor Market Outcomes, 2001-2004 Cohorts as Control Group

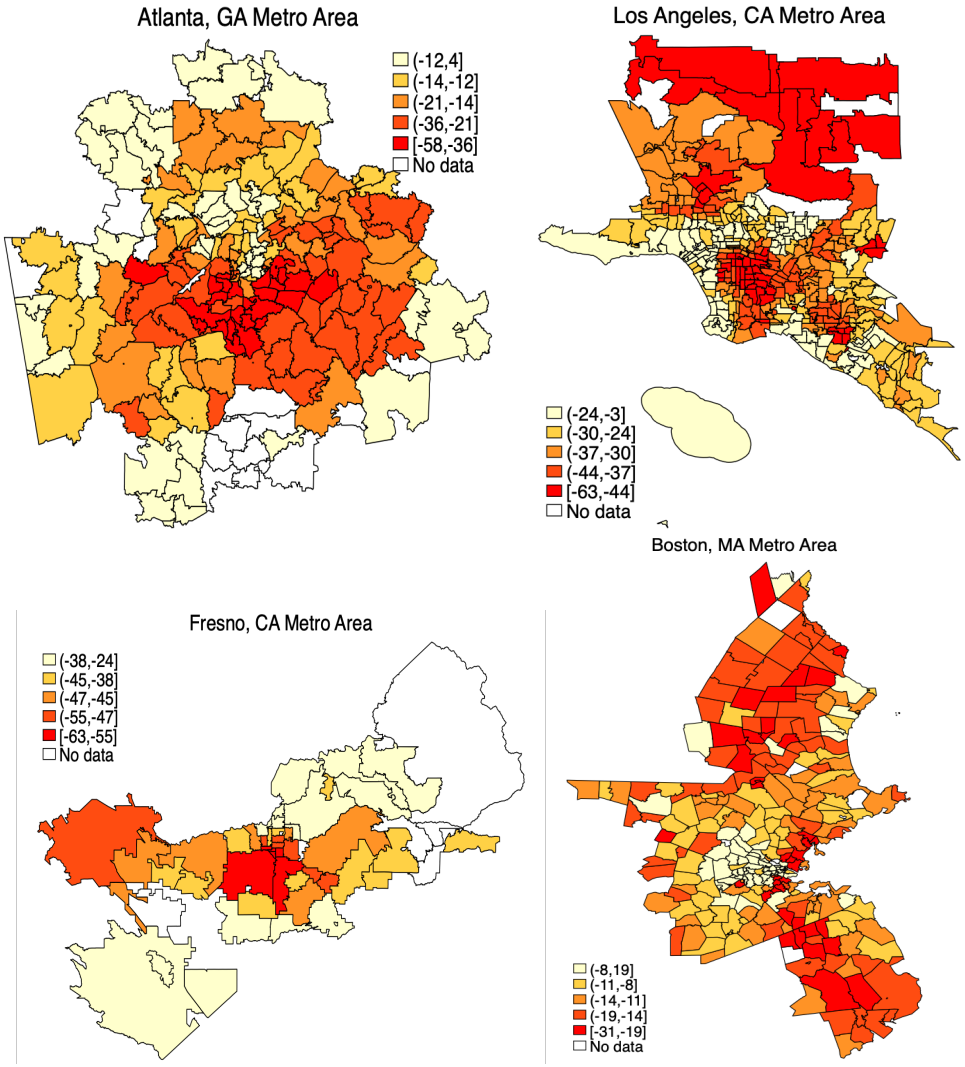
	Dependent Variable:				
	Any College	A.A. Attainment	Degree Non-completion	Current U.G. Enrollment	Current Grad Enrollment
Age 22-23	-0.00787***	-0.00241***	-0.00296***	0.00199	-0.00169**
	(0.00134)	(0.00088)	(0.00113)	(0.00127)	(0.00082)
Age 24-25	-0.00606***	-0.00173**	-0.00342***	-0.00574***	-0.00170**
	(0.00132)	(0.00087)	(0.00112)	(0.00117)	(0.00082)
Age 26-27	-0.00583***	-0.00060	-0.00502***	-0.00344***	-0.00353***
	(0.00133)	(0.00090)	(0.00115)	(0.00119)	(0.00084)
N	1194000	1194000	1194000	1194000	1194000
R^2	0.10	0.02	0.02	0.11	0.03

	Dependent Variable:				
	Any Graduate School	Earnings	Employment	Parental Co-residence	Migration Out-of-State
Age 22-23	-0.00423***	18	-0.00141	-0.00837***	0.00154
	(0.00096)	(74)	(0.00151)	(0.00151)	(0.00111)
Age 24-25	-0.00178*	279***	0.00260*	-0.01193***	0.00382***
	(0.00096)	(74)	(0.00151)	(0.00151)	(0.00110)
Age 26-27	-0.00124	354***	0.00452***	-0.01340***	0.00477***
	(0.00100)	(80)	(0.00156)	(0.00156)	(0.00115)
N	1194000	1194000	1194000	1194000	1156000
R^2	0.10	0.11	0.08	0.12	0.06

Notes: Each column displays the coefficient estimates of β_j from Equation (3), estimated using an alternate sample that uses the 2001-2004 age-18 cohorts as the control group. See Section 3.2 for the definition of the outcome variables. The ACS does not contain standard geographic information for individuals who live in group quarters, which is why the sample size for the migration outcome is slightly smaller than the other sample sizes. Standard errors are clustered by ZIP code. * if p $<$ 0.1, ** if p $<$ 0.05, *** if p $<$ 0.01.

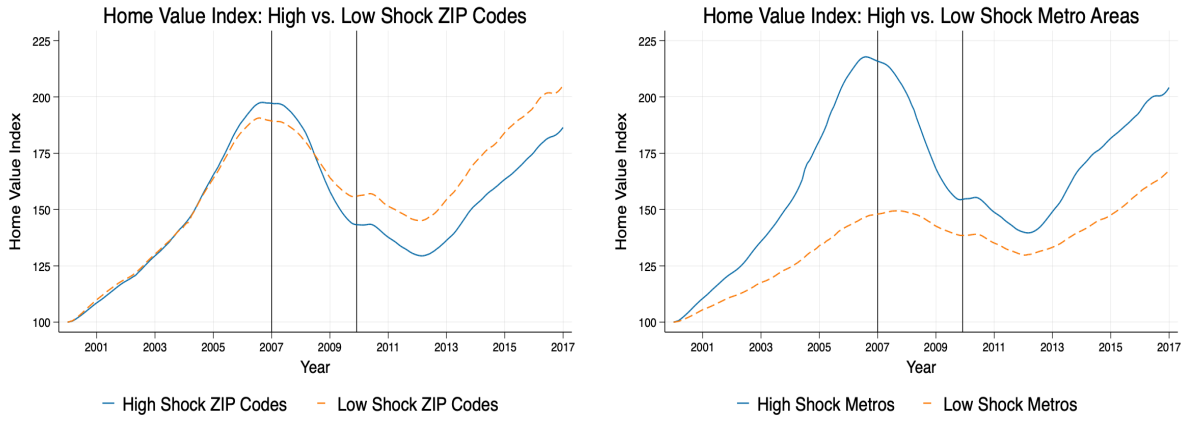
C Figures

Figure 1: Percent Change in ZIP Code Home Values Within Selected Metropolitan Areas, Jan. 2007 - Dec. 2009



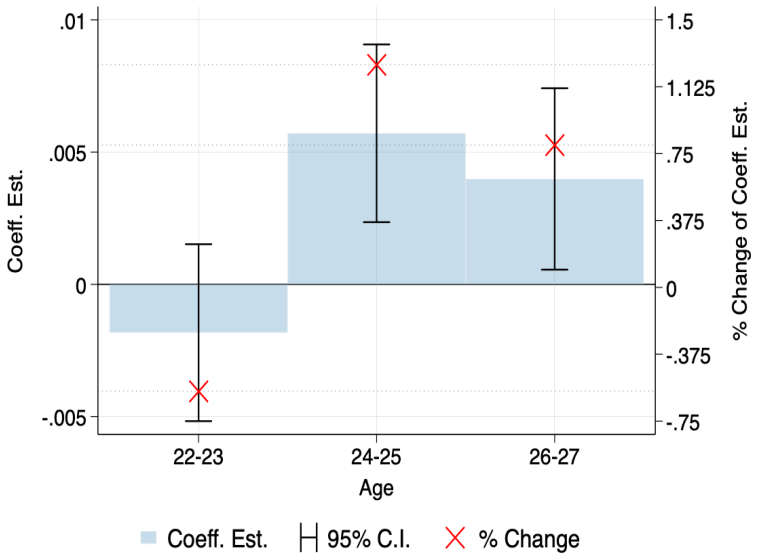
Each map displays the average percent change of house values from January 2007-December 2009 in ZIP codes within selected metropolitan areas. Categories correspond to quintiles, with darker colors representing larger decreases in ZIP code house values. Quintiles are calculated using ZIP code population in the year 2000 weights, and quintiles are calculated within each metropolitan area.

Figure 2: Housing Value Growth Within and Between Metropolitan Areas, 2000-2016



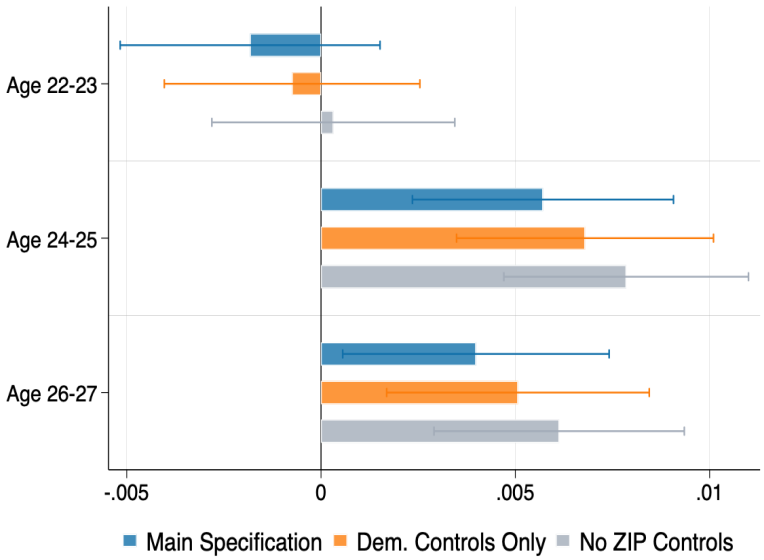
These figures display house value index growth between January 2000 and December 2016. Each figure has two time series; one representing geographies that experienced above median declines in house values during the recession, and one representing geographies that experienced below median declines in house values during the recession. The figure on the left compares house value growth of high and low shock ZIP codes within metropolitan areas. A ZIP code is considered to be a high shock ZIP code if its percent change in house values between January 2007 and December 2009 was below median (more negative). Otherwise, it is considered a low shock ZIP code if its percent change in house values between Jan. 2007 and Dec. 2009 was above median (less negative). The figure on the right compares house value growth between metropolitan areas. A metro area is considered to be a high shock metro area if its percent change in house values between January 2007 and December 2009 was below median (more negative). Otherwise, it is considered a low shock metro area if its percent change in house values between Jan. 2007 and Dec. 2009 was above median (less negative). The median is determined using population weights, and the indexes are calculated using population weights as well. The two vertical lines delineate the time period used for the main right hand side variable used in this paper, January 2007-December 2009. Data on house values used to calculate the indexes comes from the Zillow Home Value Index: `https://www.zillow.com/research/data/`.

Figure 3: Impact of Parents' House Values on Bachelor's Degree Attainment



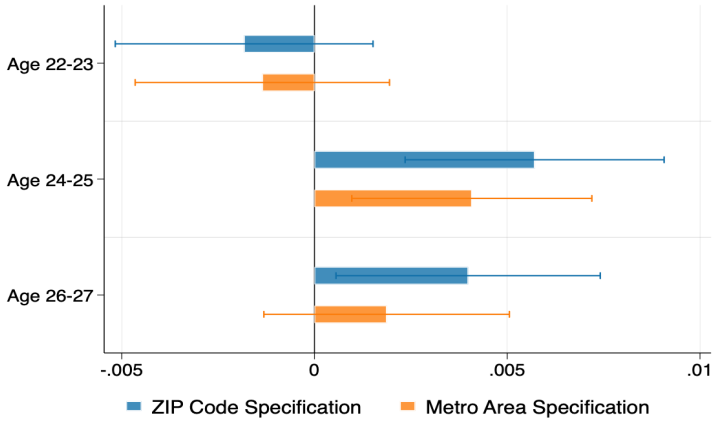
Notes: This figure reports the coefficient estimates of Equation (3) when the outcome is bachelor's degree attainment. Each bar reports the corresponding estimate of β_j, and the values are traced back to the left y-axis. These estimates are interpreted as the impact of a 10% increase of parents' house values on bachelor's degree attainment. The first bar reports an estimate of -0.00183 at age 22-23, the second bar reports an estimate of 0.00571 at age 24-25, and the third bar reports an estimate of 0.00399 at age 26-27. The Xs (red) translate the coefficient estimates into percent changes relative to the mean outcome in a given age bin. These values are traced back to the right y-axis. The left-most X reports that the estimate of β_{22-23} translates to -0.58% change in B.A. attainment, the second X reports that the estimate of β_{24-25} translates to a 1.24% change in B.A. attainment, and the right-most X reports that the estimate of β_{26-27} translates to a 0.80% change in B.A. attainment. The statistical significance of the estimates is conveyed by the inclusion of 95% confidence intervals. The estimate at age 22-23 is statistically indistinguishable from 0, while the estimates are ages 24-25 and 26-27 are statistically significant at a 95% level of confidence. Standard errors are clustered by ZIP code.

Figure 4: Estimate of Impact of Parents' House Values on B.A. Attainment by ZIP Controls



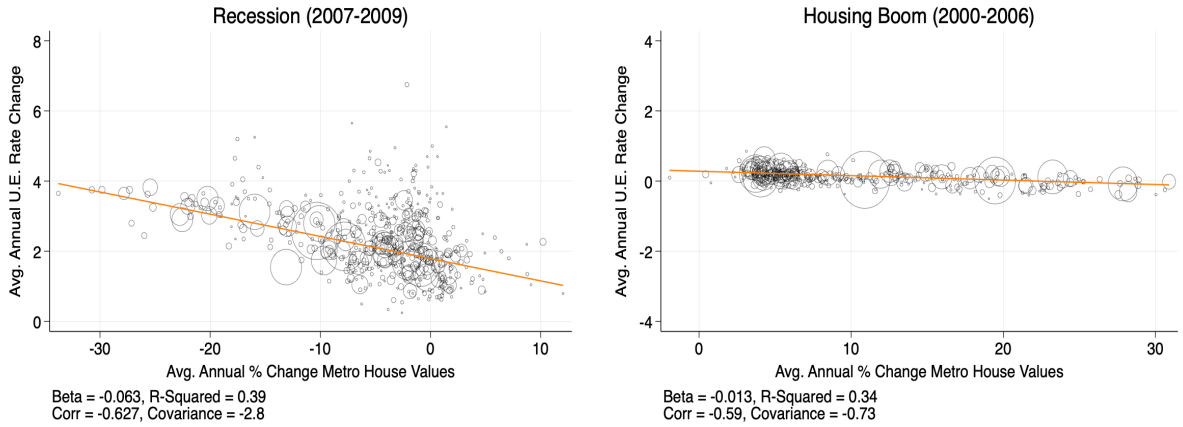
Notes: This figure reports the estimates of β_j when the outcome is bachelor's degree attainment from specifications of Equation (3) that differ by the ZIP code controls included. Within each age range, the top bar (blue) reports the estimate when all ZIP code controls are included. These are the baseline results reported in Figure 3. The middle bar (orange) reports the coefficient estimates when the only parametric ZIP Controls included are the share of residents that are black/white and share of residences with H.S. or B.A. Attainment. I refer to this specification as having demographic ZIP code controls only. The bottom bar (gray) reports estimates when no parametric ZIP controls are included. Statistical significance is conveyed by the inclusion of 95% confidence intervals. Standard errors are clustered by ZIP code.

Figure 5: Impact of Parents' House Values on B.A. Attainment: Baseline vs. Metro Specification



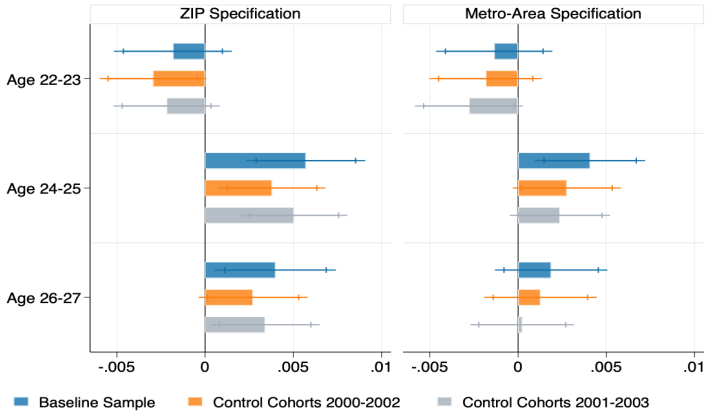
Notes: This figure compares the coefficient estimates of β_j from Equation (3) to the coefficient estimates of λ_j from Equation (4) when the outcome is bachelor's degree attainment. Within each age bin, the top bar (blue) reports the coefficient estimate from Equation (3), and the bottom bar (orange) reports the coefficient estimates of Equation (4). Statistical significance is conveyed by the inclusion of 95% confidence intervals. Standard errors are clustered by ZIP code when estimating Equation (3), and are clustered by metropolitan area when estimating Equation (4).

Figure 6: Comparing the Relationship Between Local Unemployment Rates and Local House Value Changes During the Great Recession and the 2000s Housing Boom



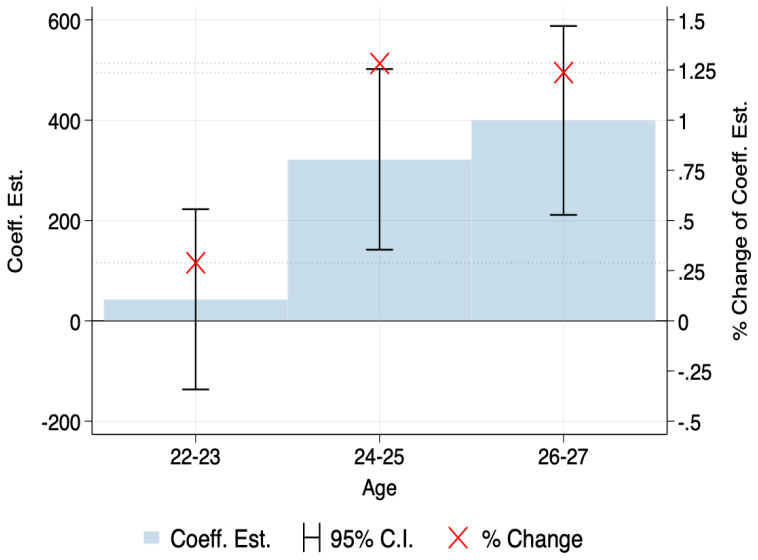
Notes: These two figures display the results of a weighted regression of metropolitan area average annual unemployment rate change on metropolitan area average annual percent change in house values. The figure on the left reports results for the Great Recession years of 2007-2009, and the figure on the right reports results for the pre-recession Housing Boom years of 2000-2006. The variable on the y-axis is the average annual change in a metropolitan area's unemployment rate. The variable on the x-axis the average annual percent change in a metropolitan area's house values, measured by Zillow's ZHVI. The coefficient estimate during the recession is -0.063, implying that a 1% increase in housing values decreases the unemployment rate by 0.063 p.p.. The coefficient estimate during the housing boom is -0.013, implying that a 1% increase in metro-area housing values decreases local unemployment rates by 0.013 p.p.. The R^2 from each regression is also reported in the figures, along with the correlation and covariance between the two variables.

Figure 7: Robustness of Results to Alternative Control Cohorts



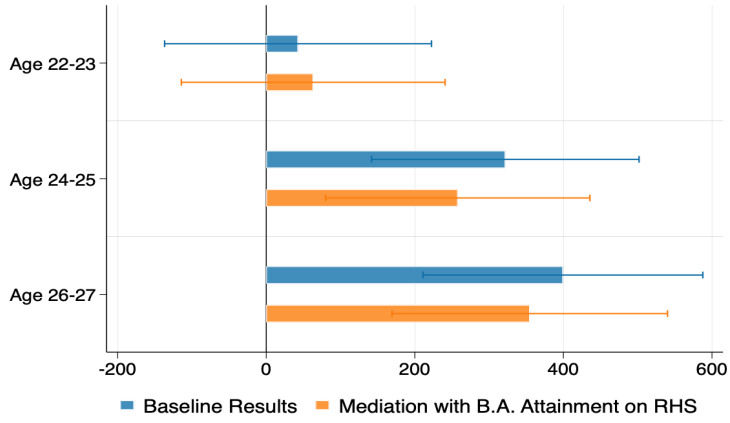
Notes: This figure reports coefficient estimates of β_j from Equation (3) and λ_j from Equation (4) that use alternative cohorts in the control group. The left panel reports the coefficient estimates from Equation (3) (ZIP Specification) and the right panel reports the coefficient estimates from Equation (4) (Metro-Area Specification). The top bar (blue) in each age bin reports the coefficient estimates using the baseline sample. The middle bar in each bin (orange) reports coefficient estimates when the control group includes cohorts turning 18 in the years 2000-2002. The bottom bar in an age bin (grey) reports coefficient estimates when the control group includes cohorts turning 18 in the years 2001-2003. Statistical significance is conveyed by the inclusion of bars that show a 90% and 95% confidence interval. Standard errors are clustered by ZIP code in the ZIP Specification and by Metropolitan Area in the Metro-Area Specification.

Figure 8: Impact of Parents' House Values on Annual Earnings



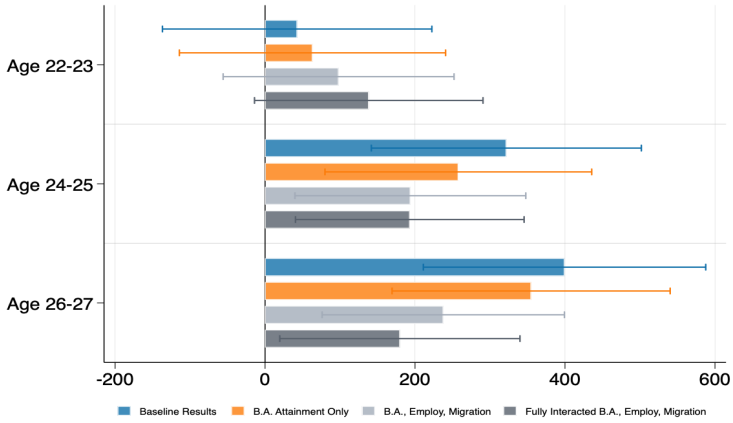
Notes: This figure reports the coefficient estimates of Equation (3) when the outcome is annual earnings. Annual earnings are measured in 2011 $s. Each bar reports the corresponding estimate of β_j, and the values are traced back to the left y-axis. These estimates are interpreted as the impact of a 10% increase of parents' house values on annual earnings. The first bar reports an estimate of $42.81 at age 22-23, the second bar reports an estimate of $321.8 at age 24-25, and the third bar reports an estimate of $399.4 at age 26-27. The Xs (red) translate the coefficient estimates into percent changes relative to the mean outcome in a given age bin. These values are traced back to the right y-axis. The left-most X reports that the estimate of β_{22-23} translates to 0.29% change in B.A. attainment, the second X reports that the estimate of β_{24-25} translates to a 1.28% change in B.A. attainment, and the right-most X reports that the estimate of β_{26-27} translates to a 1.24% change in B.A. attainment. The statistical significance of the estimates is conveyed by the inclusion of 95% confidence intervals. The estimate at age 22-23 is statistically indistinguishable from 0, while the estimates are ages 24-25 and 26-27 are statistically significant at a 95% level of confidence. Standard errors are clustered by ZIP code.

Figure 9: Mediating the Impact of Bachelor's Degree Attainment on Earnings



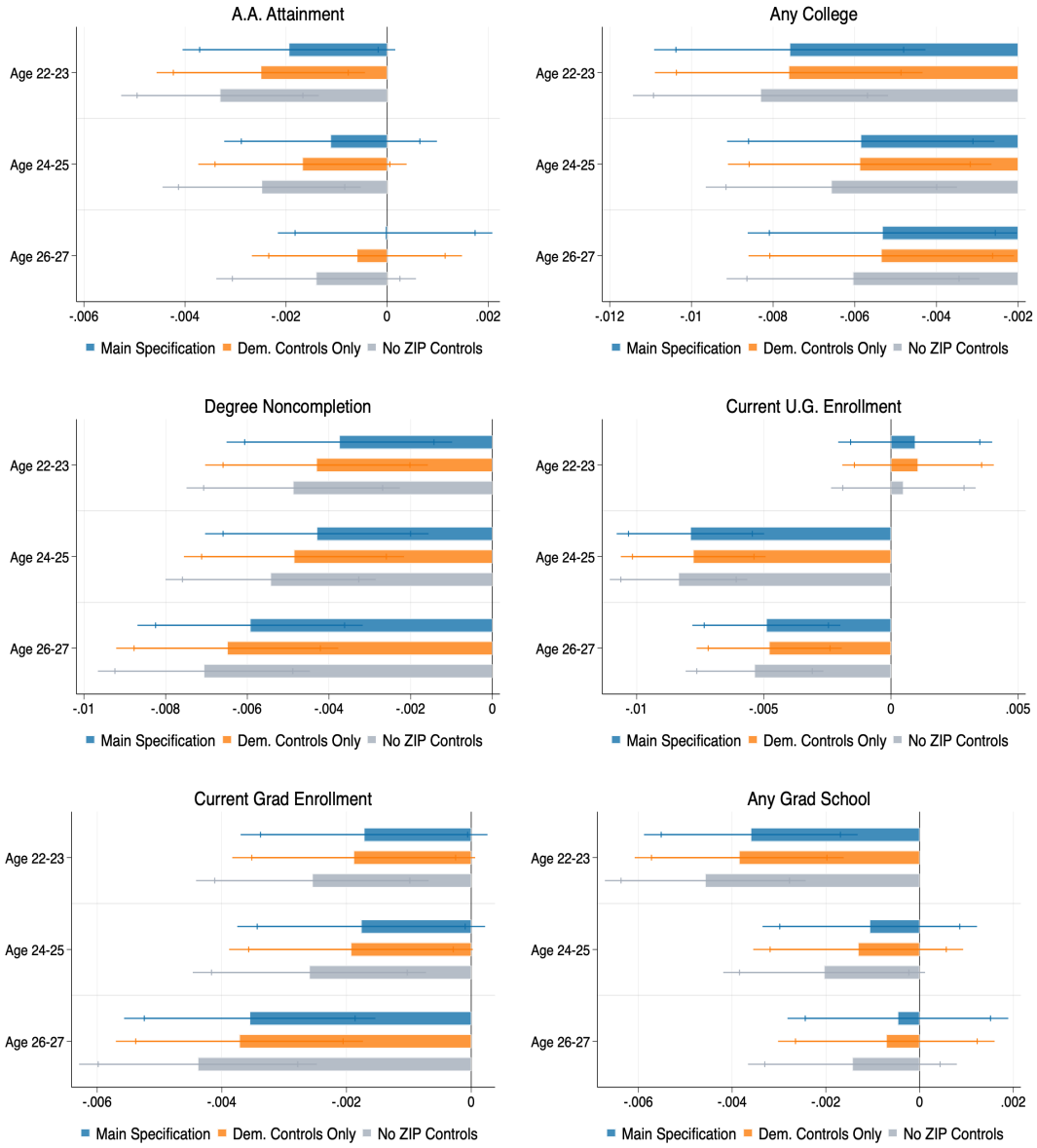
Notes: This figure compares the baseline coefficient estimates of β_j from Equation (3) when the outcome is earnings to the coefficient estimates of a version Equation (3) where the indicator for bachelor's degree attainment is added as an independent variable. This latter specification is the mediation exercise. Within each age bin, the top bar (blue) reports the baseline coefficient estimate from Equation (3), and the bottom bar (orange) reports the coefficient estimates from the mediation exercise. Statistical significance is conveyed by the inclusion of 95% confidence intervals. Standard errors are clustered by ZIP code.

Figure 10: Mediating the Impact of Parents' House Values on Earnings with B.A. Attainment, Employment, and Migration



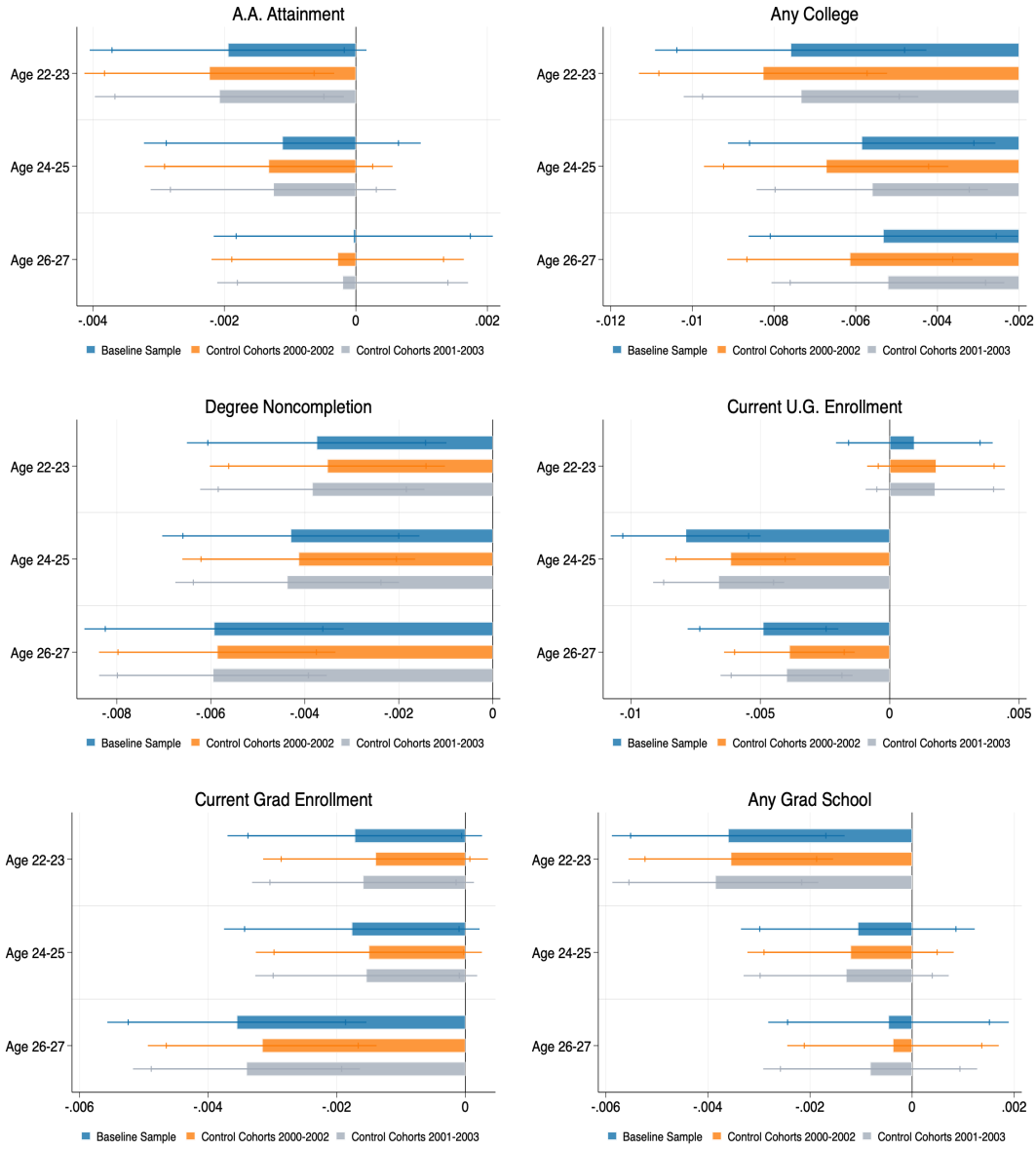
Notes: This figure compares the baseline coefficient estimates of β_j from Equation (3) to coefficient estimates from three separate mediation exercises. The first exercise adds bachelor's degree attainment as a control to Equation (3). The second adds B.A. attainment, full-time employment, and out-of-state migration as control variables to Equation (3). The final exercise fully interacts B.A. attainment, full-time employment, and out-of-state migration and adds them Equation (3). In each age bin, the top bar (blue) reports the baseline coefficient estimate from Equation (3). The second bar from the top (orange) reports the coefficient estimates from the mediation exercise that only adds B.A. attainment. The second bar from the bottom (light grey) reports the coefficient estimates from the mediation exercise that adds B.A. attainment, employment, and migration as controls to Equation (3). The bottom bar (dark grey) reports coefficient estimates from the mediation exercise in which B.A. attainment, migration, and employment are fully interacted and added to Equation (3). Statistical significance is conveyed by the inclusion of 95% confidence intervals. Standard errors are clustered by ZIP code.

Figure C.1: Estimates of Parents' House Values on Education Outcomes by ZIP Controls



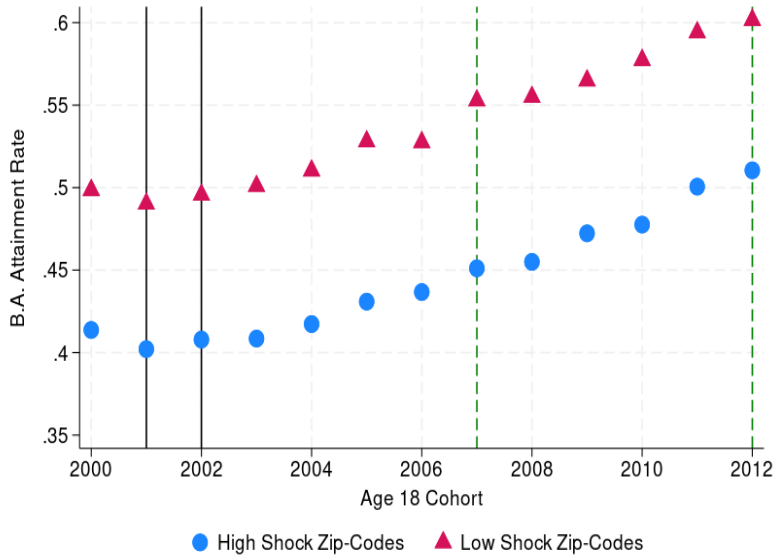
Notes: This figure reports the estimates of β_j when for various college education outcomes from specifications of Equation (3) that differ by the ZIP code controls included. Within each age range, the top bar (blue) reports the estimate when all ZIP code controls are included. These are the baseline results reported in Table 2. The middle bar (orange) reports the coefficient estimates when the only ZIP Controls included in Equation (3) are share of residents that are black/white and share of residences with H.S. or B.A. Attainment. I refer to this specification has having demographic ZIP code controls only. The bottom bar (gray) reports the estimates of Equation (3) when no ZIP controls are included. Statistical significance is conveyed by the inclusion of 90% and 95% confidence intervals. Standard errors are clustered by ZIP code.

Figure C.2: Robustness of College Education Results to Alternative Control Cohorts



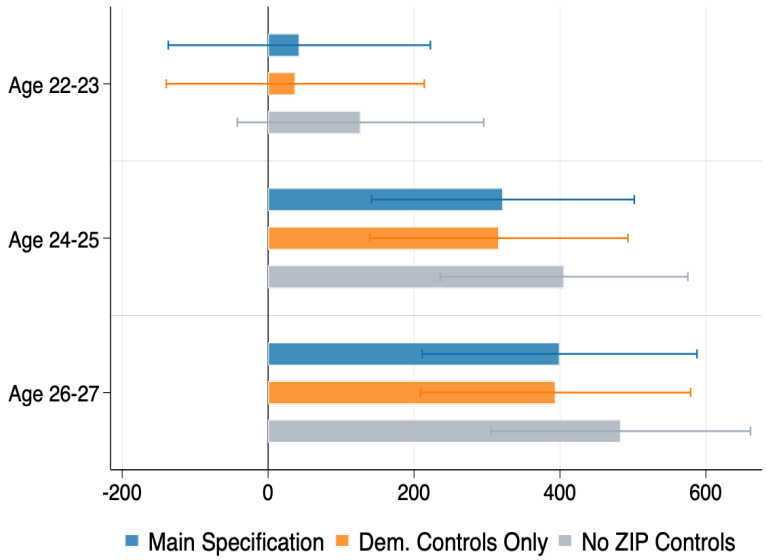
Notes: This figure reports coefficient estimates of β_j from Equation (3) that use alternative cohorts in the control group. The top bar (blue) in each age bin reports the coefficient estimates using the baseline sample. The middle bar in each bin (orange) reports coefficient estimates when the control group includes cohorts turning 18 in the years 2000-2002. The bottom bar in an age bin (grey) reports coefficient estimates when the control group includes cohorts turning 18 in the years 2001-2003. Statistical significance is conveyed by the inclusion of bars that show a 90% and 95% confidence interval. Standard errors are clustered by ZIP code.

Figure C.3: Cohort Trends in Age 26-27 Bachelor's Degree Attainment by Type of ZIP Code



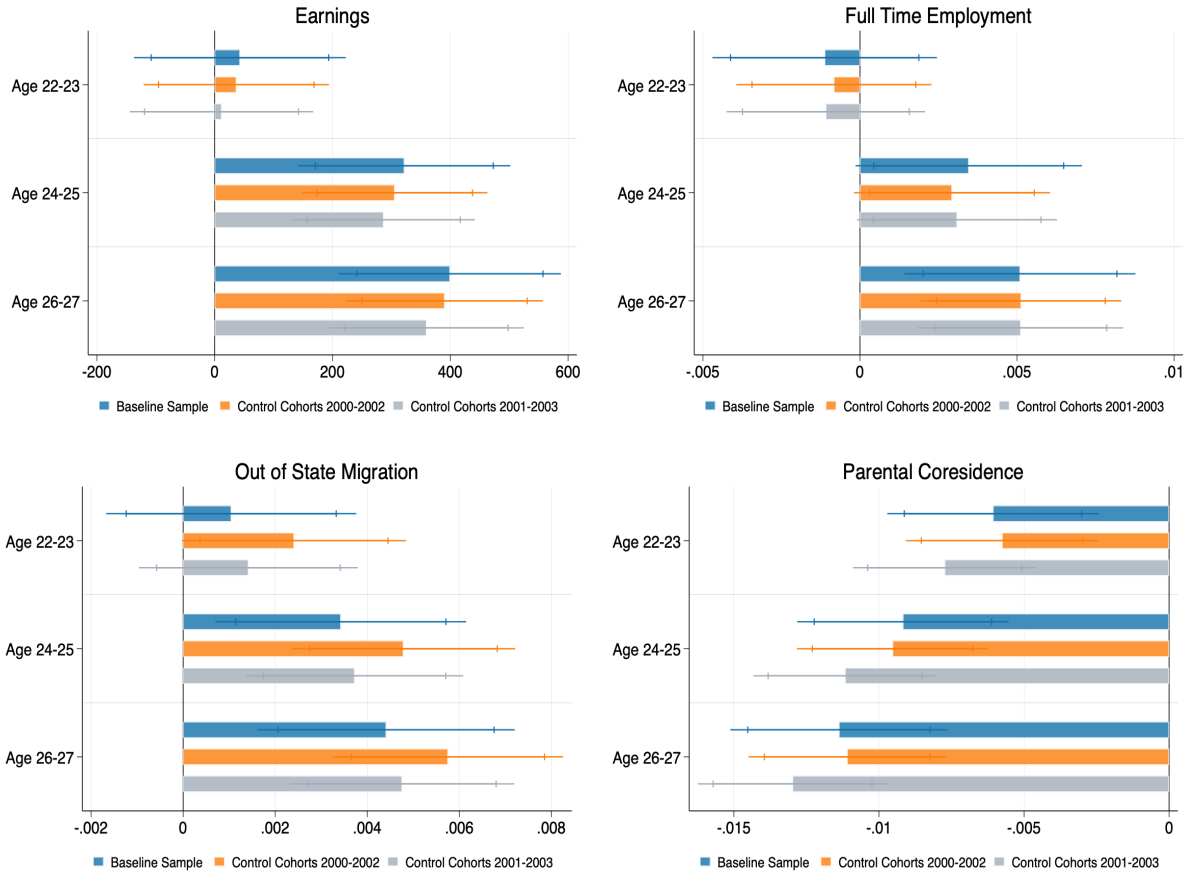
Notes: This figure displays the average B.A. attainment rate at ages 26-27 of individuals who grew up in ZIP codes that experienced large vs. small recession shocks to house values. High shock ZIP codes (represented by the blue dots) are ZIP codes that experienced above-median (more negative) changes to ZIP code house values between January 2007-December 2009 within their metropolitan area. Low shock ZIP codes (represented by the red triangles) are ZIP codes that experienced below-median (less negative) changes to ZIP code house values between January 2007-December 2009 within their metropolitan area. There are high and low shock ZIP codes within each metropolitan area. Each dot or triangle represents the B.A. Attainment rate for a given cohort. Cohorts are indexed by the year an individual turns 18 years old. The two solid lines on the left (black) delineate the control cohorts in the main analysis sample, while the two dashed lines (green) delineate the treatment cohorts in the main analysis sample. The main takeaway from this figure is that children who grow up in ZIP codes experiencing larger (more negative) shocks to house values during the recession have lower levels of bachelor's degree attainment. Data used to construct this figure comes from the ACS, MAF-ARF, 2000 Decennial Census, and the Zillow Home Value Index.

Figure C.4: Estimates of Impact of Parents' House Values on Annual Earnings by ZIP Controls



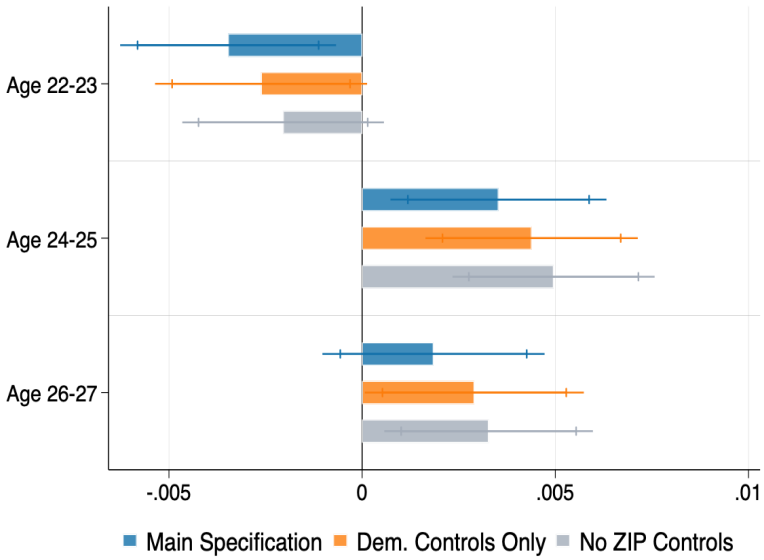
Notes: This figure reports the estimates of β_j when the outcome is earnings from specifications of Equation (3) that differ by the ZIP code controls included. Within each age range, the top bar (blue) reports the estimate when all ZIP code controls are included. These are the baseline results reported in Figure 3. The middle bar (orange) reports the coefficient estimates when the only ZIP Controls included in Equation (3) are share of residents that are black/white and share of residences with H.S. or B.A. Attainment. I refer to this specification has having demographic ZIP code controls only. The bottom bar (gray) reports the estimates of Equation (3) when no ZIP controls are included. Statistical significance is conveyed by the inclusion of 95% confidence intervals. Standard errors are clustered by ZIP code.

Figure C.5: Robustness of Labor Market Results to Alternative Control Cohorts



Notes: This figure reports coefficient estimates of β_j from Equation (3) that use alternative cohorts in the control group. The top bar (blue) in each age bin reports the coefficient estimates using the baseline sample. These results are reported in Figure 8 and Table 3. The middle bar in each bin (orange) reports coefficient estimates when the control group includes cohorts turning 18 in the years 2000-2002. The bottom bar in an age bin (grey) reports coefficient estimates when the control group includes cohorts turning 18 in the years 2001-2003. Statistical significance is conveyed by the inclusion of bars that show a 90% and 95% confidence interval. Standard errors are clustered by ZIP code.

Figure C.6: Estimate of Impact of Parents' House Values on B.A. Attainment by ZIP Controls Using the 2001-2004 Age-18 Cohorts as the Control Group



Notes: This figure reports the estimates of β_j when the outcome is bachelor's degree attainment from specifications of Equation (3) that differ by the ZIP code controls included. Within each age range, the top bar (blue) reports the estimate when all ZIP code controls are included. These are the baseline results reported in Figure 3. The middle bar (orange) reports the coefficient estimates when the only ZIP Controls included in Equation (3) are the share of residents that are black/white and the share of residences with H.S. or B.A. Attainment. I refer to this specification has having demographic ZIP code controls only. The bottom bar (gray) reports the estimates of Equation (3) when no ZIP controls are included. All estimates reported use the 2001-2004 age-18 cohorts as the control group. Statistical significance is conveyed by the inclusion of 95% confidence intervals. Standard errors are clustered by ZIP code.